

Cohort Seventeen >>

Top Twenty Five Finalist Selection

Customer Energy Savings

Grid Assets

Materials

Grid Management

Grid Protection

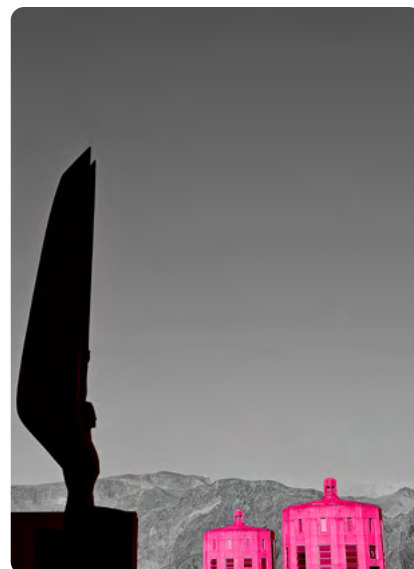
The Importance of Networks >>

JULY 13th, 2026

More than three years ago, the team behind the Carolinas Engine for Grid Modernization (CEGM) set out to persuade the National Science Foundation (NSF) that the Carolinas were worthy of a third Engine – this time focusing on advancing the pace of innovation and commercial deployment of grid technologies.

The NSF Engines program was established to “*advance transdisciplinary, collaborative, and use-inspired and translational research and technology development in key technology focus areas.*” Engines can receive up to \$160 million to “*support the development of regional coalitions of researchers, institutions and companies to conduct research and development that engages people in the process of creating solutions with economic and societal impacts.*”

Last summer we were notified that the CEGM was selected as one of 29 semifinalists to advance to the next stage of review by the NSF. In September we received confirmation that the CEGM had been selected as one of just 15 finalists from the original pool of more than 300 applicants – and to prepare for a site visit from the NSF leadership.



On February 23rd, UNC Charlotte hosted nearly 70 energy and grid leaders from across the region and beyond to support the site visit by the NSF, and to field industry specific questions. Joules Accelerator had the privilege to be a part of this team, and to help shape the Engine’s translation and commercialization narrative. As part of this effort, Joules called upon its network – and the network answered. We invited nearly a dozen board members, ecosystem partners, and past startups to join us. People who believe in our mission traveled from Chicago, Atlanta, Washington, D.C., and Raleigh to make it happen. For that we are deeply grateful.

Today we eagerly await the decision from the NSF. Regardless of outcome, the CEGM preparation, team building, and ecosystem outreach brought into focus the critical importance of networks – and how these networks can accelerate innovations from lab to pilot to scale.

As you review our Cohort 17: Top 25 Selection Book, please take time to read “The Grid Can See Now” by Mila Hunt, P.E., Strategy Office Director at GE Vernova and “What Utilities Actually Look for in a Pilot” by Ernst Scholtz, CTO at S&C Electric Company. On July 27th and 28th, we will convene for the Future Grid 26 event where we will discuss innovation strategies, opportunities for collaboration, and advanced grid priorities.

We hope you will join us.

Sincerely,

Ryan Rutledge
Executive Director, Joules Accelerator

Table of Contents

Introduction

A letter from the Joules Team	2
Cohort 17 Recap & Trends	4
The Grid Can See Now - Will it Act?	5
What Utilities Actually Look for in a Pilot	6
Future Grid 26 Event Series	7
Program News & Announcements	7

Top 25

Selection Instructions	8
Overviews	9
Executive Summaries	14
 Customer Energy Savings	14
 Grid Assets + Materials	19
 Grid Management	25
 Grid Protection	35

Thank You

Fellows & Associates	39
----------------------	----

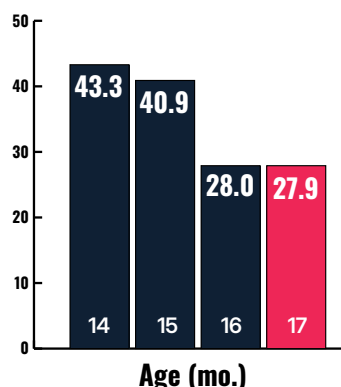
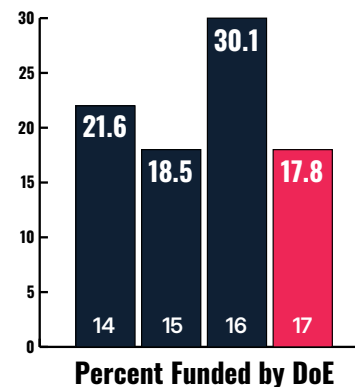
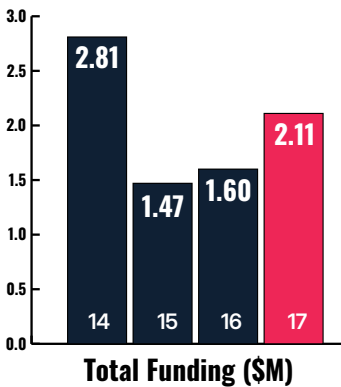
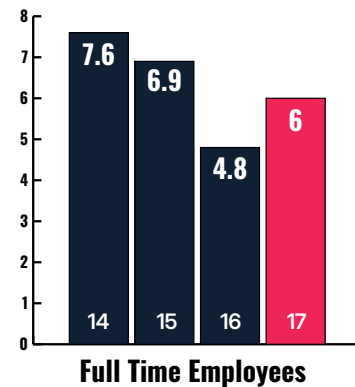
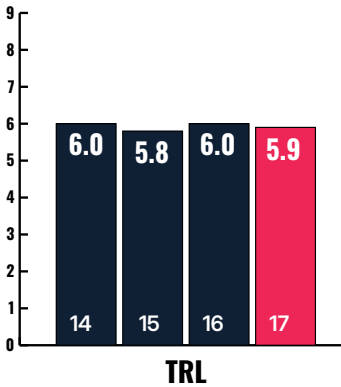
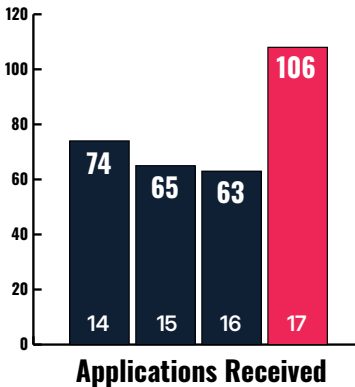


Cohort 17 Recap & Trends

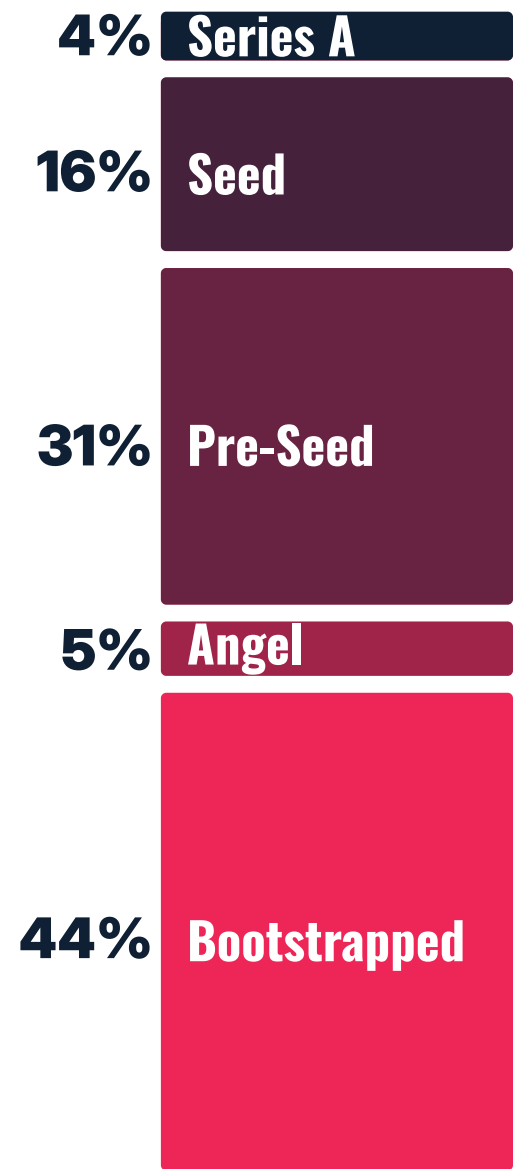
106 Applications from 13 Countries

As compared to the past three cohorts, this application cycle revealed:

- There is *more interest than ever* in the Joules Accelerator program.
- Applicant TRL remains steady at ~6, but average age is down. *Startups are moving faster.*
- Early-stage startups are *doing more with less.* More bootstrapping and fewer FT employees.
- Total funding raised by startups is still off from the highs of 2023 & 2024, *but trending up again.*
- Applicant funding from DoE is *down sharply.*



Applicant Stage of Funding



A comparison across Cohorts 14, 15, 16, and 17.

The Grid Can See Now

The question is whether it will act

Contributed by **Mila Hunt, P.E.**, Strategy Office Director at GE Vernova



The grid is the glaring and compounding bottleneck to economic development globally. Interconnection queues are measured in years. Substation operators are running assets decades past retirement. Outages are driven by violent non-conforming weather events that have long outpaced original design limits. These are present day operating realities.

Now layer in the AI data center supercycle. City-scale loads are arriving faster than any planning cycle can absorb. Power demand swinging hundreds of megawatts within seconds from hyperscale compute bursts is a fundamentally new demand signature on infrastructure that was engineered for a different physics.

What makes this moment particularly curious is the tension at its center: the same capital wave stressing the grid is beginning to fund its transformation.

The data center buildout is the largest demand shock the power sector has absorbed in a generation, and it is doing something that decades of internal utility planning discussions could not: creating urgency. Hyperscalers and neoclouds are forcing three structural gaps into the open. The interconnection queue vacuum, where power procurement timelines now rival transmission build cycles. The behind-the-meter orchestration void, where campus-scale loads operate without the real-time control architecture their volatility demands. And a chronic shortage of grid engineering expertise capable of specifying, designing, and commissioning the infrastructure this moment requires.

Capital is now chasing all three. Not cleanly or without friction, but at a scale and velocity the grid has not seen before.

The undercurrent I think matters most in this phase of the energy transformation is the convergence happening at two ends of the grid simultaneously.

At the asset level, visibility is arriving. Sensors, monitoring devices, and edge intelligence are finally making the grid observable in ways it has never been before. Half the physical equipment on the average utility network is at or past end of life. For too long, operators have been flying partially blind, making maintenance decisions based on time intervals rather than asset condition, learning about failure reactively instead of predictively. Condition-based monitoring,

line sensors, transformer diagnostics, and wide-area phasor measurement are changing that equation. The data exists. The challenge is turning it into actions and decisions.

At the system level, edge automation is maturing. Zonal controllers, adaptive protection, distributed intelligence, and grid-forming capabilities are moving from pilots to operational deployments. Grid topology is increasingly delivering on promised dynamic, bidirectional operation that must be orchestrated in real time, with millions of new endpoints and inverter-based resources that contribute no inertia and behave nothing like the synchronous generators the grid was designed around. This is now fully, in all its complexity, the living reality for power delivery operators.

The convergence of asset visibility and edge automation is where the real prize lives. When you can see what every asset is doing and take local corrective action at millisecond speed, the grid stops being reactive and starts being resilient by design. That is what a modernized grid looks like. It's not rebuilt in the image of the old one, but one where sensing, communication, edge control, and analytics work as a layered stack, continuously learning and adapting.

For startups building in this space, my observation is this: the utilities that matter most are not asking for pilots. They are asking for scalable, interoperable, rate-case-defensible solutions that can be deployed without disrupting the system they are already running. We're overhauling the car while racing down the autobahn. Winning in this market requires deep domain credibility and dependability with deep knowledge of the physics and operating environment. Speaking the customer's language before you walk in the door is commercial leverage.

The AI data center wave did not create the grid's problems. But it is creating the urgency and mobilizing the capital to finally fix them. That is the opportunity for everyone building at the grid edge today.



Mila Hunt, P.E. leads Grid Automation Strategy at GE Vernova, shaping its digital substation, protection and control, and grid automation software portfolio, including Grid-Beats™. Previously at Energy Impact Partners, she worked with entrepreneurs, utilities, and investors to commercialize novel grid technologies across North America. She holds a degree from UT Austin's McCombs School of Business.

What Utilities Actually Look for in a Pilot

How startups can get it right

Contributed by Ernst Scholtz, Chief Technology Officer at S&C Electric Company

Every year, the number of promising grid technologies emerging from vendors and startups exceeds what utilities can realistically evaluate. Based on experience, the distance between a strong demo and a successful utility pilot is wider than most founders expect, and it has little to do with whether the technology works. As a company that has spent more than a century focused on delivering innovations for the electric grid and that works closely with both utilities and emerging technology vendors, S&C sits at an interesting vantage point in this dynamic. A few patterns from that experience are worth sharing with this year's Joules cohort.

Utilities are evaluating innovation, not just invention

A dazzling invention helps in creating an initial buzz for a startup, but it does not necessarily guarantee a successful product. When a utility evaluates a pilot, it is then ascertaining whether the proposed solution can operate reliably and with resiliency under real operating conditions, not just in a controlled environment. To make that jump is where true innovation occurs: transforming a promising idea into a solution that performs in the real world.



Demonstration of S&C's portfolio of Overhead and Underground solutions for a reliable and resilient distribution grid.

The unglamorous criteria matter most

Pilots that move successfully toward deployment rarely win on novelty alone. They win on fit: does the solution integrate with existing equipment and communications, does it meet the utility's security and operational requirements, and can it be maintained by existing field crews without adding complexity? Utilities remember solution providers who showed up prepared, communicated clearly about what

the technology could and could not yet do, and made the collaboration easier rather than harder.

Scope the pilot to answer a specific question

The most effective pilots are built around a single, well-defined question the utility cares about, rather than a broad showcase of everything technology can do. A startup that asks a utility to evaluate open-ended potential puts the burden of defining success on the utility. A startup that comes in with a specific, measurable hypothesis, framed around outage impact, restoration time, or another outcome the utility already tracks, gives the team something concrete to assess and to advocate for internally.

Engage the whole organization - not just one champion

Pilots often begin because one engineer or innovation lead is excited about a new technology. But a pilot's path forward depends on people across the organization, including operations, field crews, and procurement, who were not necessarily part of the early conversation. Engaging them early, even informally, often determines whether a promising pilot moves forward or stalls.

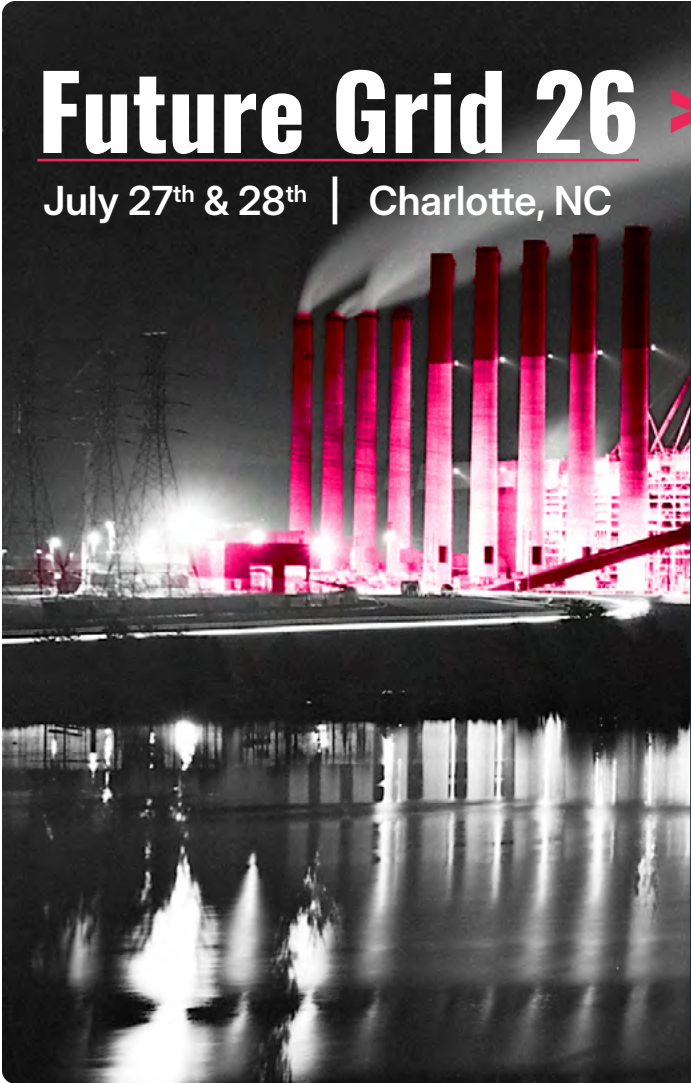
Working alongside an established player can help

This is part of why we increasingly see startups succeed by working alongside an established company like S&C. Leveraging decades of utility relationships, field-proven equipment, and an understanding of how the grid actually operates can shorten a startup's path to a meaningful pilot, allowing the startup to focus its energy on what it does best: the innovation itself.

The utility environment rewards patience, precision, resilience, and a clear understanding of what a single pilot can prove. Accelerators like Joules exist because the industry needs pathways for innovative ideas to make it into practice. The startups that succeed are the ones that understand that earning a utility's trust is itself part of the product.



Ernst Scholtz is S&C Electric Company's Chief Technology Officer. He leads the company's Technology and Break-through Innovation (TBI) team, which focuses on technology-led innovations, knowledge creation, and prototyping of future offerings. Before joining S&C, Ernst spent 20 years at Hitachi Energy, formerly ABB, first as a scientist, then leading the corporate research organization.



Future Grid 26 >>

July 27th & 28th | Charlotte, NC

Hosted by: Duke Energy, Honeywell, and EY

Guest Keynote from Dr. Vanessa Chan

Featuring speakers and panelists from:
 Aligned Climate Capital - Streetlife Ventures
 Clean Energy Ventures - Dominion Energy
 Innovation Center - Halliburton Labs - Catalyzer
 Ventures - S&C Electric Company - GE Vernova
 UNC Charlotte - Hitachi Energy - Xcel Energy
 Duke Energy - Southern Company - Georgia
 Power - National Grid Partners - and more!

Startup pitches from:
 Magnefy - Solitude Labs - Scenergy - Every
 Electric - Causergy - EcoPhi - Path Power - Pila

Click Here to Register Today
Space is Limited

Program News & Announcements

Events, Funding Announcements, and Board Updates



Program News

- 

Jaideep Malik, Senior Manager at EY Parthenon joined the Joules BoD in Q2. Jaideep brings decades of energy innovation and finance experience to the program. Welcome aboard!
- 

Lili Zay joined the Joules team full-time in Q1 2026 as the Director of Operations and Communications. Lili is a Charlotte native, Morehead-Cain Scholar, and conservation leader. Welcome, Lili!
- 

Charlotte, NC - Jan. 21st: Joules launched the first day of Future Grid 26 with the R&D Showcase. UNC Charlotte's EPIC and Duke Energy hosted more than 75 guests to dive into energy and grid research.
- 

Atlanta, GA - Mar. 2-4, 2027: After sponsoring the Initiate Startup Program at DTECH 26, Joules returns as the anchor sponsor in 2027. Come see our startups pitch in Atlanta. [Register now.](#)

Cohort Funding Announcements

- 

Neara (Cohort 5) announced an AUD \$90M Series D raise led by TCV on February 9th, 2026. The round will support expanding grid capacity. ([Announcement](#))
- 

DG Matrix (Cohort 13) announced a \$60M Series A round led by Engine Ventures and joined by ABB Ventures on February 18th, 2026. ([Announcement](#))
- 

Electric Era (Cohort 11) received a \$5.05M grant from Washington State DoT to expand EV charging capacity across the state. ([Announcement](#))
- 

WeaveGrid (Cohort 4) announced a partnership with General Motors Energy to support access to utility programs across the country. ([Announcement](#))
- 

Asset Cool (Cohort 14) opened its new Terrawatt-1 in Batley June 12th. The facility expands capacity to meet rapidly growing demand. ([Announcement](#))
- 

Solitude Labs (Cohort 15) raised \$425,000 in angel funding from undisclosed investors on March 27th, 2026. ([Announcement](#))

Selection Instructions

Help select the next cohort of six startups

Step 1: Review the Top 25

Please review the Top 25 summaries on the following pages. Take a deeper dive on any startups that you find interesting by reading the one-page application. Please reach out if you have questions or comments about any of the startups, technologies, or applications.

Step 2: Think Like a Founder

These startups are looking for connections to pilots and customers. Think about how you or your organization can support startups with these opportunities.

Step 3: Make Your Selection

Click the link below and share your Cohort 17 selections with us.

**Click Here to Record
Your Selections**



Cohort 17 Selections due by midnight on Sunday, July 26th!

Syncrowin, Inc

San Francisco, CA - United States

"Syncrowin provides autonomous Industrial AI Autopilot for Energy."



Page 14

>> [Link to Website](#)

PoolFlex

San Francisco, CA - United States

"PoolFlex enables pools to participate in energy grid events."



Page 15

>> [Link to Website](#)

Edgecom Energy

Toronto, ON - Canada

"Edgecom provides an AI-powered platform helping C&I facilities cut energy costs, reduce emissions, and earn grid revenue."



Page 16

>> [Link to Website](#)

FLUIX Inc.

San Francisco, CA - United States

"FLUIX deploys autonomous AI models that optimize and unlock capacity in data centers."



Page 17

>> [Link to Website](#)

Raya Power, Inc.

San Francisco, CA - United States

"Raya Power makes savings, resilience, and VPP participation possible for the 80% of Americans locked out of traditional solar."



Page 18

>> [Link to Website](#)

Vertical Semiconductor

Somerville, MA - United States

"Vertical Semi is developing vertical GaN power transistors to build the foundational power layer for the next era of compute."



Page 19

>> [Link to Website](#)

DexMat

Houston, TX - United States

"DexMat manufactures Galvorn, an advanced carbon material giving engineers a copper alternative at a fifth the weight."

DEXMAT

Page 20

>> [Link to Website](#)

Boreal Conductors

Montreal, QC - Canada

"We transform superconducting tapes to improve the resilience and robustness of superconducting power devices."



Page 21

>> [Link to Website](#)

Bedrock Semiconductor

Hummelstown, PA - United States

"We are developing semiconductor devices to increase the energy efficiency of electrical systems."



Page 22

>> [Link to Website](#)

Sonder Labs

Solna - Sweden

"Sonder Labs is developing power balancing technology for critical infrastructure that the world cannot afford to switch off."

 Sonder Labs

Page 23

>> [Link to Website](#)

Scaled Power

Arlington, VA - United States

"Scaled Power provides coaxial power electronics to reduce the footprint and deployment time of power infrastructure for datacenters and utilities."



Page 24

>> [Link to Website](#)

RunPilot

Sheffield - United Kingdom

"RunPilot helps utility planning teams route and compare infrastructure options before delivery risks become costly."



Page 25

>> [Link to Website](#)

Pravāh

San Francisco, CA - United States

"Pravāh provides ML for the grid: helping utilities plan hardening, speed up interconnection studies, and cut outages."



Page 26

>> [Link to Website](#)

Cloneable AI

Raleigh, NC - United States

"Cloneable uses AI to capture field utility data and automate the back-office engineering process around it."



Page 27

>> [Link to Website](#)

Axolotis.ai, Inc.

Durham, NC - United States

"Axolotis unlocks untapped energy in existing hydroelectric assets through digital twins, optimization, and AI."



Page 28

>> [Link to Website](#)

Enoda Ltd.

Edinburgh, Scotland - United Kingdom

"ENODA technology enables utilities to unlock additional capacity from existing grid infrastructure."



Page 29

>> [Link to Website](#)

Watchfully Inc.

Cary, NC - United States

"Watchfully fuses live agentic power flow with AI vision inspection to reveal capacity, mitigate risk & speed interconnection."



Page 30

>> [Link to Website](#)

CubeNexus

Chicago, IL - United States

"CubeNexus unifies industrial and energy data across space and time using AI agents and a cube-based spatial data model."



Page 31

>> [Link to Website](#)

Holdpoint

North Sydney, NSW - Australia

"Holdpoint provides energization proof for mission-critical infrastructure."



Page 32

>> [Link to Website](#)

KanopyAI

New York, NY - United States

"Kanopy helps utilities convert field video into 3D maps of grid conditions, risks, and work needs."



Page 33

>> [Link to Website](#)

Tirion Systems

Buenos Aires - Argentina

"Tirion provides an AI-native physical observability platform for the 100,000 invisible utilities the enterprise stack forgot."



Page 34

>> [Link to Website](#)

OmniSpectra

Dallas, TX - United States

"OmniSpectra helps utilities prioritize transformer maintenance before failures occur with affordable sensing."



Page 35

>> [Link to Website](#)

Orbital Sentry

Washington, DC - United States

"Orbital Sentry detects wildfires within seconds of ignition via continuous infrared monitoring from geosynchronous orbit."



Page 36

>> [Link to Website](#)

Firescape

Albuquerque, NM - United States

"Firescape provides wildfire risk mitigation software to help utilities prevent catastrophic wildfire."



Page 37

>> [Link to Website](#)

GridIQ

Clayton, CA - United States

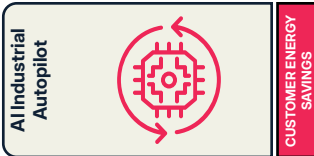
"GridIQ's patent-pending hardware and diagnostics platform deliver real-time fault detection, localization, and classification."



Page 38

>> [Link to Website](#)

SYNCROWIN



Company Overview

Point of Contact Information

Name: Aisshwarya Kansakaar
Title: CEO and Co-Founder
Email: ash@syncrowin.com.au
Phone: +61 401741158

Company Information

Legal Name: Syncrowin, Inc
Address: 350 California Street, 4th Floor, San Francisco, CA 94104
Date Founded: 1/24/2024
Web Address: <https://www.syncrowin.com>
Number of Employees: 5FT 1PT

Founding Team

Founders & Leadership

Name: Aisshwarya Kansakaar
Title: CEO and Co-founder
Email: ash@syncrowin.com.au



Name: Jaideep Upadhyay
Title: CTO and Co-founder
Email: jai@syncrowin.com.au

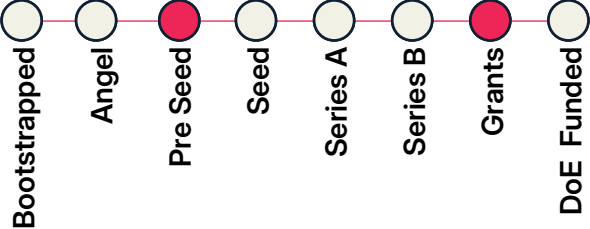


Funding & Traction

Revenue

2024: N/A
2025: \$150,000
2026 proj.: \$600,000

Latest Round of Funding



Funding Raised to Date: \$2,400,000

Grants, Contracts, Memberships

CSIRO RISE Accelerator Grant – US\$135,000, Melbourne Accelerator Program (MAP) – A\$20,000, non-dilutive funding
Microsoft- US\$ 150,000 in builder credits; Multiple commercial paid pilots with leading utilities, IPPs and renewable asset owners in APAC;

TAM, Tech, & Supplemental

Total Addressable Market

\$328 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Renewable energy operators face increasing grid complexity, intermittent generation, aging assets, labour shortages, and fragmented operational systems. Control rooms rely heavily on manual monitoring and reactive decision-making across multiple software platforms. Our ideal customers are utilities, IPPs, ISOs, grid operators, and renewable asset owners managing large portfolios of solar, wind, BESS, and hybrid energy assets.

Value Proposition

Syncrowin Industrial AI Autopilot acts as a 24/7 intelligent operations decision intelligence lead. It continuously monitors assets, predicts operational issues before they occur, recommends optimal actions, and helps operators maximise generation, improve reliability, reduce downtime, lower O&M costs, and make faster, data-driven decisions using existing infrastructure.

Product or Service

Syncrowin provides an enterprise AI Autopilot system for renewable energy operations. The platform integrates with SCADA, EMS, historians, weather feeds and enterprise systems to deliver AI-powered forecasting, real-time monitoring, predictive maintenance, anomaly detection, dispatch optimisation, through a single operational layer integrated with existing SCADA systems delivered as enterprise SaaS.

Founding Team Description

Built by engineers and energy operators, our founding team combines 20+ years of experience across AI, renewable energy, industrial systems and digital twins. CEO Ash has led enterprise AI and automation build and deployments across global manufacturing and industrial sectors, while CTO Jaideep Upadhyay holds multiple patents and has delivered 70+ digital twin and critical infrastructure projects in Australia.

Technologies & Unique Know-How

Syncrowin combines proprietary AI, time-series forecasting, computer vision, anomaly detection, optimisation engines, digital twins into a unified operational intelligence system. Our models are purpose-built for renewable energy control rooms, enabling continuous operational awareness, predictive insights and operational decision support while integrating seamlessly with existing utility infrastructure.

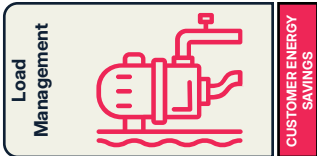
Traction & Competitive Landscape

Syncrowin has deployed its AI with leading renewable energy companies across Australia and India, supporting more than 1.6 GW of renewable assets. Unlike traditional monitoring systems or standalone AI tools, Syncrowin provides a unified AI Control Room system that combines monitoring, forecasting, optimisation, and operator decision support in one enterprise offering, reducing operational complexity and improving asset performance.

Pilot Demonstration & Cost

A minimum 20-22 week demonstration with a utility, IPP, or renewable energy owner minimum 100-500 MW portfolio of solar, wind, or BESS assets. Syncrowin integrates with existing SCADA, historian, EMS and weather systems to deploy AI for generation forecasting, operational decision optimisation, asset performance optimisation, anomaly detection, and operator decision support. Cost: \$135,000- \$170,000

POOLFLEX



Company Overview

Point of Contact Information

Name: William Rodriguez
Title: Founder and CEO
Email: william@getpoolflex.com
Phone:

Company Information

Legal Name: PoolFlex
Address: 33 W 17th St floor 6, New York, NY 10011
Date Founded: 04/17/2026
Web Address: <https://getpoolflex.com/>
Number of Employees: 1FT

Founding Team

Founders & Leadership

Name: William Rodriguez Jimenez
Title: Founder and CEO
Email: william.a@alum.mit.edu



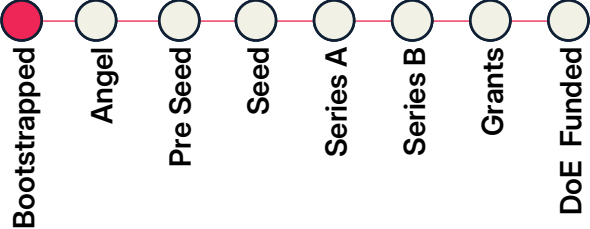
Name:
Title:
Email:

Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: N/A

Latest Round of Funding



Funding Raised to Date:

Grants, Contracts, Memberships

We participated in the University of Oxford's OX1 Incubator. We are currently participating in Antler VC's Founder Residency in New York City.

TAM, Tech, & Supplemental

Total Addressable Market

\$300 million

Tech Readiness Level

- ①
- ②
- ③
- ④
- ⑤
- ⑥
- ⑦
- ⑧
- ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

Grid operators fear how they are going to supply the huge wave of 30% demand growth in the next decade. Generation and interconnection queues are stalled for years. Whether through load curtailment or additional generation, utilities are willing to throw money at whatever can help them lower their burden. Our ideal customers and economic buyers are utilities who want to invest to lower their energy delivery burdens. On the other side are pool owners who want to lower their energy bills.

Value Proposition

PoolFlex enables smart controls for pool-related energy loads to unlock grid resilience. We do this by aggregating pool-related energy loads (over 2GW in the US) and working with utility demand response programs to deliver 1) savings and 2) cash payments to the pool owners.

Product or Service

We sell grid resilience to utilities while producing a passive income stream for pool owners.

Founding Team Description

William decided to innovate the energy industry after witnessing the energy disaster following Hurricane Maria in his native Puerto Rico. Since then, William has developed a professional background at the intersection of energy and technology. William has degrees from MIT and Oxford, with research focusing on EV charging and energy forecasting. He's worked at Tesla, Iberdrola, Apple, Morgan Stanley, the United Nations and startups. He's on a mission to make energy more reliable and affordable.

Technologies & Unique Know-How

Our core technology is a load-aggregation and dispatch platform built on two areas our competitors lack in combination: (1) deep equipment-level integration with pool systems, including thermal modeling that lets us stagger curtailment across a site without compromising water quality; (2) industry experience at Tesla, Iberdrola, and other energy companies where William led aggregator registrations. We know how to do this for other energy assets, so we can do this for pools and then expand.

Traction & Competitive Landscape

We are early-stage, currently in Antler's NYC residency. We are closing in founding clients (gyms, bathhouses, spas). We are targeting a summer 2027 launch. We are in talks with major sunbelt utilities. Our competitors are horizontal aggregators (Voltus, EnergyHub, CPower) that have not tapped in this space and OEMs (Pentair, Hayward) that own hardware but do not want to become energy companies. Our edge is pool-specific controls and curtailment guarantees in a segment too niche for incumbents.

Pilot Demonstration & Cost

Our ideal partner is Duke Energy or a similarly large utility investing in demand response. First, we would test our product on 5-10 commercial facilities (gyms, spas, aquatic centers), integrating pumps, heaters, and chillers. Second, we'd recruit a cohort of homeowners with variable-speed pumps. Overall success metrics would include measured kW curtailment on the utility-side and comfort/water-quality impact on the asset-owner side.
Cost: \$75,000

EDGECOM ENERGY



C&I Energy Management

CUSTOMER ENERGY SAVINGS

Company Overview

Point of Contact Information

Name: Tanner Behrend
Title: VP Strategic Growth
Email: jboyce@edgecom.ai
Phone: (647) 821-4416

Company Information

Legal Name: Edgecom Energy Inc.
Address: Toronto, Ontario

Date Founded: 01/01/2016
Web Address: <https://edgecom.ai/>
Number of Employees: 30FT

Founding Team

Founders & Leadership

Name: Behdad Bahrami
Title: CEO & Co-Founder
Email: bbahrami@edgecom.ai



Name: Mehdi Parvizi
Title: CTO & Co-Founder
Email: mparvizi@edgecom.ai

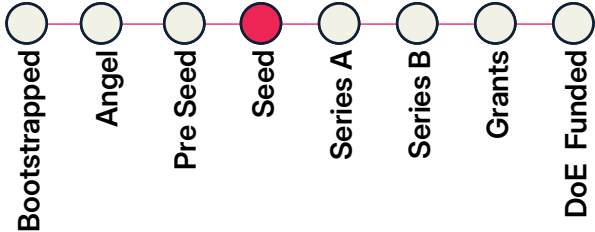


Funding & Traction

Revenue

2024: ~\$3.4M
2025: ~\$4.5 M
2026 proj.: ~\$7.4M

Latest Round of Funding



Funding Raised to Date: \$2.8M

Grants, Contracts, Memberships

NRC IRAP, SR&ED tax credit, FedDev Ontario- Federal Economic Development Agency for Southern Ontario, Sustainable Development Technology Canada (SDTC) Decentralized Energy Canada (DEC)

TAM, Tech, & Supplemental

Total Addressable Market

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Commercial and industrial facilities hold significant flexible load, but most lack the visibility, automation, and market access needed to participate in grid programs. Edgecom solves this with AI-powered energy orchestration that reduces energy costs, lowers emissions, and enables automated grid participation. Our ideal customers are multi-site C&I operators such as manufacturers, cold storage facilities, data centres, commercial real estate portfolios, and EV fleet operators.

Value Proposition

Edgecom transforms idle building flexibility into revenue through an AI-powered energy orchestration platform that requires no upfront capital or operational disruption. Our platform predicts grid conditions, autonomously dispatches assets, and delivers verified demand response performance. Customers reduce peak energy costs, unlock new grid revenue streams, and access wholesale energy markets without needing a dedicated energy management team.

Product or Service

Edgecom's AI Energy CoPilot combines three products: dataTrack™ for real-time IoT energy monitoring, pTrack® for AI-driven peak forecasting, and NeuraCharge™ for autonomous DER dispatch. Together, these modules help commercial and industrial facilities reduce energy costs, automate demand response participation, and generate grid revenue by intelligently controlling HVAC, batteries, generators, and other flexible assets.

Founding Team Description

Edgecom's founding team combines deep expertise in energy management, AI, and grid operations. CEO Behdad Bahrami built Edgecom after firsthand experience managing complex industrial energy systems. CTO Mehdi Parvizi, PhD, developed the platform's forecasting and dispatch algorithms. Together, the leadership team has scaled Edgecom to 256+ facilities, 1+ GW of managed load, and \$350M+ in customer savings, with extensive utility, DER, and market experience.

Technologies & Unique Know-How

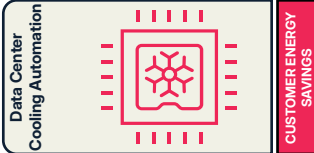
Edgecom uniquely combines real-time load monitoring, AI-driven grid forecasting, and autonomous DER dispatch in one platform. Unlike competitors focused only on demand response, building controls, or analytics, Edgecom orchestrates entire facilities across multiple asset types. Our hardware-agnostic platform delivers utility-grade M&V and is powered by years of facility and grid performance data, enabling pTrack® forecasting accuracy of 99.1%.

Traction & Competitive Landscape

Edgecom is a commercially deployed, utility-validated platform with 256+ facilities, 1+ GW of managed load, and \$350M+ in customer savings. The company generated \$4.5M USD in 2025 revenue with 88% YoY growth and manages ~71.8 MW in Ontario's IESO capacity auction. Unlike competitors focused on single solutions, Edgecom combines AI forecasting, facility-wide orchestration, and utility-grade M&V in one integrated platform deployed at regulatory scale.

Pilot Demonstration & Cost

A small-scale pilot would deploy Edgecom's AI Energy CoPilot at 5–10 commercial and industrial facilities with controllable load assets such as HVAC, batteries, refrigeration, or EV charging. The pilot would include IoT submetering, AI forecasting calibration, and live automated demand response dispatch events with utility-grade M&V reporting. An ideal partner is a utility or grid operator seeking scalable demand flexibility and DER orchestration for C&I customers.
Cost: \$43,000



Company Overview

Point of Contact Information

Name: Abhishek Sastri
Title: CEO
Email: Abhi@fluix.ai
Phone: (407) 801-2078

Company Information

Legal Name: FLUIX Inc.
Address: 1160 Battery Street East, STE 100,
San Francisco, CA 94111
Date Founded: 11/18/2021
Web Address: <https://www.fluix.ai/>
Number of Employees: 2FT 2PT

Founding Team

Founders & Leadership

Name: Abhishek Sastri
Title: CEO
Email: Abhi@fluix.ai



Name: Chase Overcash
Title: CTO
Email: Chase@fluix.ai

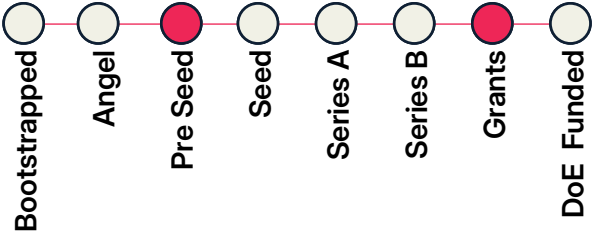


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$109,500

Latest Round of Funding



Funding Raised to Date: \$2,300,000

Grants, Contracts, Memberships

Venture 4 Climatetech, TechConnect AI for Grid Resilience, Cleantech Open; TVA EnergyRight, TenHats (data center), 7-site latam colocation; Techstars Industries of the Future, Pi Labs Growth Programme, Morgan Stanley Inclusive Ventures, etc.

TAM, Tech, & Supplemental

Total Addressable Market

\$12 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

AI data centers are running into power limits. Operators need more tokens per MW, but today cooling, power, and compute infrastructure are optimized in silos with static controls. Our ideal customer is a power-constrained AI, colo, enterprise, or utility-backed data center that needs more useful compute from existing infrastructure without risking uptime.

Value Proposition

FLUIX helps data centers produce more tokens per MW by applying on-prem reinforcement learning across the physical AI infrastructure stack. Cooling is the first wedge because it is controllable today, but the platform expands into power orchestration, grid flexibility, and compute-aware infrastructure control, without touching tenant workloads or replacing equipment.

Product or Service

We sell A.I.M.I.®, an on-prem AI control platform for data center infrastructure. A.I.M.I. connects to existing BMS, EPMS, workload management, HVAC, meters, and sensors. It starts read-only, learns site behavior, then runs guarded autonomous control with operator override, failsafes, and M&V reporting.

Founding Team Description

FLUIX is led by founders with rare overlap across data centers, thermal systems, and reinforcement learning. Abhi is a second-time founder who previously developed patented liquid cooling technology for data centers and worked on DoD mission-critical thermal/data center systems. Chase has built RL models across healthcare, transportation, and energy in research labs at UCI, ASU, and related programs.

Technologies & Unique Know-How

FLUIX combines on-prem reinforcement learning, physics-aware controls, digital twin methods, and protocol-level integration with existing facility systems. Our edge is safely applying RL to real data centers with guardrails, fallbacks, audit logs, and no cloud dependency for control. We are building the control layer for cooling, power, and compute infrastructure.

Traction & Competitive Landscape

FLUIX is live in 3 data center environments, has signed its first recurring license, and is moving toward 9 sites in motion through utility-backed deployments and customer expansions. A public TVA/TenHats case study showed 6.6% PUE improvement with no new sensors or SLA excursions. Competitors focus on BMS, DCIM, or HVAC optimization; FLUIX is positioned as an on-prem RL control layer for AI infrastructure.

Pilot Demonstration & Cost

FLUIX has already proven autonomous cooling optimization in live data centers, helping unlock more power headroom. The next pilot tests A.I.M.I.® for power orchestration: using existing meters, ATS/generators, cooling data, and utility pricing/grid signals to recommend when a site can safely shift from grid to onsite power, reduce peak costs, support demand response, and maintain uptime. We already have data center and utility partners ready to test this. Cost: \$25,000-\$75,000 depending on site complexity and scope.

RAYA POWER



Company Overview

Point of Contact Information

Name: Ciara Corbeil
Title: Head of Bus. Dev. and Operations
Email: ciara@rayapower.com
Phone:

Company Information

Legal Name: Raya Power, Inc.
Address: 34 Buchanan Street, Apt 12, San Francisco, CA, 94102
Date Founded: 2/8/2024
Web Address: <https://rayapower.com/>
Number of Employees: 3FT 5PT

Founding Team

Founders & Leadership

Name: Meghan Wood
Title: Co-Founder and CEO
Email: meghan@rayapower.com

Name: Nicole Gonzalez
Title: Co-Founder and CTO
Email: nicole@rayapower.com

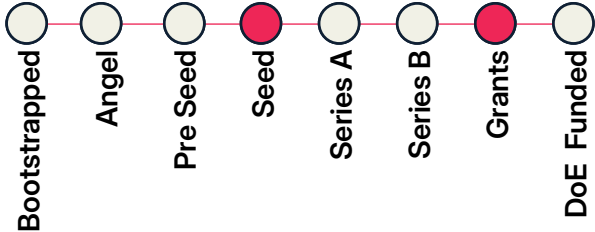


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$400,000

Latest Round of Funding



Funding Raised to Date: \$1,940,000

Grants, Contracts, Memberships

Stanford Impact Founder Fellowship, Walking Softer, Parallel18 & Pre-18 at Puerto Rico Science, Tech, and Research Trust, PYMES Innovadoras; >\$11M in LOIs, \$240K pre-orders, \$60K contract with the Environmental Defense Fund, Discussions with two major US utilities regarding a DER pilot

TAM, Tech, & Supplemental

Total Addressable Market

\$250 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

US electricity costs are up 40% since 2021 (PowerLines) and outages are frequent and increasing, potentially rising 100x by 2030 (DOE). Raya serves the ~80% of Americans locked out of traditional rooftop solar—renters, low-to-middle-income single family homeowners, and anyone without the budget, timeline, or roof integrity for a \$25k+ installation—and utilities building DER capacity. We are currently focused on single-family or small multi-family homes with ground space for the system.

Value Proposition

- Resilience AND affordability for underserved customers: Delivers back-up (with recharge) for critical loads AND bill savings; - No back-feeding, easy set up: No interconnection, permits, wiring changes, or home modifications needed. Set up in <3 hours; - Demand response: Can reduce household grid draw in response to utility signals; - Appliance-level monitoring and control: Load-level prioritization for customers, performance insights for appliance OEMs, planning and load management for utilities.

Product or Service

Raya Power’s modular, integrated solar+battery systems power critical appliances (e.g., refrigerators, fans, WiFi, etc.). Systems can optimize energy sources and loads to maximize outage readiness, cost savings, customer appliance needs, and grid capacity. Systems set up on the ground in <3 hours, no permitting, grid interconnection, or home modifications required.

Founding Team Description

Nicole Gonzalez (CTO) brings a background in Electrical Engineering from Princeton, an MS in Design Impact Engineering from Stanford, and experience engineering NASA’s Mars Rover - systems-level design and test under extreme reliability constraints, directly applicable to field-resilient energy hardware. Meghan Wood (CEO), Forbes 30 Under 30, holds a Stanford MBA and MS in Renewable Energy - giving the team direct energy-market and finance fluency.

Technologies & Unique Know-How

Raya Power provides savings and resiliency to a customer segment that is locked out of traditional rooftop solar and battery backup.

Our system allows device-level visibility and prioritization, optimizes between solar, battery, and grid power, supports VPP programs without interconnection, and recharges itself, all in an easy to set-up and beautifully designed package. It generates novel appliance level data that can be monetized to appliance OEMs.

Traction & Competitive Landscape

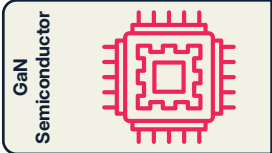
- Rooftop solar + storage: Costs \$25k+ vs. Raya’s ~\$6.7k starting price, requires homeownership, 6–12 months, permits and structural changes; - Balcony / plug-in solar: Offers savings but not convenient resilience (shuts off during outages). Faces US regulatory barriers; - Portable battery + solar panel kits: Provides backup but no integrated generation or meaningful bill savings, not designed for daily use; - Generators: Must configure each time for backup, fuel costs, health/emissions risks, noise

Pilot Demonstration & Cost

Raya proposes a 10-unit pilot with a utility partner, deploying solar-plus-battery home energy systems to vulnerable, low-income customers in high-outage-risk zones. The pilot tests on-bill financing for day-one customer savings and validates appliance-level demand response through relay-controlled load shifting. Success builds the case for a larger utility-backed deployment program.

Cost: \$85,000

VERTICAL SEMICONDUCTOR



GRID ASSETS + MATERIALS

Company Overview

Point of Contact Information

Name: Cynthia Liao
Title: CEO & Co-Founder
Email: cynthia@verticalsemi.com
Phone: (203) 299-6452

Company Information

Legal Name: Vertical Semiconductor
Address: 444 Somerville Ave Somerville, MA 02143
Date Founded: 10/24/2024
Web Address: www.verticalsemi.com
Number of Employees: 9FT 6PT

Founding Team

Founders & Leadership

Name: Cynthia Liao
Title: Co-Founder & CEO
Email: cynthia@verticalsemi.com

Name: Dr. Joshua Perozek
Title: Co-Founder & CTO
Email: josh@verticalsemi.com

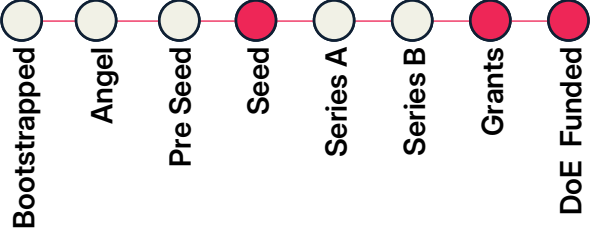


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: N/A

Latest Round of Funding



Funding Raised to Date: \$11,275,000

Grants, Contracts, Memberships

ARPA-E; Massachusetts Tech Collaborative: PROPEL grant, Massachusetts Clean Energy Center: Catalyst and DICES grant; 17 customers on a waitlist for engineering samples, developing partnerships with integrators to align device roadmap with emerging MW data center rack power requirements.

TAM, Tech, & Supplemental

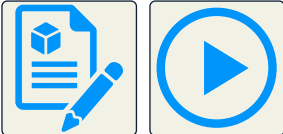
Total Addressable Market

\$24 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

As data center power moves to 800V DC distribution and 1MW rack design, Vertical Semiconductor is addressing a critical bottleneck in modern AI compute systems: the inability of existing semiconductor technologies to simultaneously deliver high voltage, high power density, and low losses. Our ideal customers are data center power system designers, power module makers, and utility-scale infrastructure suppliers building 800 VDC and 1500 VDC power architectures.

Value Proposition

Vertical Semi's vertical gallium nitride (GaN) devices operate up to 2.3 kV with 2–5× higher power density and ~40% lower losses than conventional approaches, cutting power infrastructure footprint by ~50%. Replacing state-of-the-art SiC with vertical GaN cuts energy loss by a third and saves over 50% of space, reducing cooling load and lowering total cost of ownership by an estimated 15%.

Product or Service

Vertical Semiconductor is commercializing a patented, MIT-invented vertical GaN FinFET power transistor for high-voltage, high-current conversion. GaN's wide bandgap and high mobility enable smaller, lighter, more efficient power electronics, shrinking magnetics and cooling systems. As a fabless device company, we design and sell discrete power transistors to power systems developers for data centers, energy infrastructure, and industrial power applications.

Founding Team Description

Cynthia Liao (CEO) brings expertise in energy infrastructure, finance, climate policy, and emerging technology deployment. Dr. Joshua Perozek (CTO) holds a PhD from MIT, where he developed our vertical GaN technology. Prof. Tomás Palacios (Advisor) is an IEEE Fellow and Director of MIT's Microsystems Technology Laboratories. The team pairs deep device-physics know-how with commercialization experience, backed by \$11M in seed funding from leading deep tech investors.

Technologies & Unique Know-How

GaN semiconductors switch faster and with far lower energy loss than silicon, enabling smaller, more efficient power systems. Lateral GaN devices can't reach the voltages data centers need. Vertical GaN is the only architecture that combines lateral GaN's switching speed and efficiency and scales from 40 V to 2.3 kV matching competing silicon carbide on voltage. We're taking a fabless approach using mature silicon tools, opening up opportunity to leverage mature manufacturing capability.

Traction & Competitive Landscape

Vertical has 17 customers across data center and high-power markets signed up to test engineering samples, and is collaborating with system integrators to align specs, packaging, and production toward megawatt-class racks. We outcompete silicon carbide on efficiency and power density and lateral GaN on voltage, making vertical GaN the only technology that delivers high voltage, high efficiency, and high density at once. Competitors include Infineon, Wolfspeed, Inno-science, onsemi, and Navitas.

Pilot Demonstration & Cost

A focused first pilot would deploy our 1.2 kV vertical GaN devices on the secondary (800 VDC) side of a solid-state transformer (SST), where efficient DC conversion feeds the data center rack. This contained, high-value stage proves low-loss switching at 800 VDC in real infrastructure and scales into full SST and rack-level deployment. ABB or GE Vernova would be ideal partners, pairing our devices with their power systems and grid-scale deployment reach. Cost: Single stage pilot is ~\$500k.

Company Overview

Point of Contact Information

Name: Asena Hertz

Title: CMO

Email: asena@dexmat.com

Phone: (978) 944-6811

Company Information

Legal Name: DexMat

Address: Houston, TX, 77082

Date Founded: 12/20/2019

Web Address: <https://dexmat.com/>

Number of Employees: FT11 PT1

Founding Team

Founders & Leadership

Name: Dmitri Tsentlovich, PhD

Title: Co-Founder & CTO

Email: dmitri@dexmat.com



Name: Matteo Pasquali, PhD

Title: Co-Founder & CSA

Email: matteo@dexmat.com



Funding & Traction

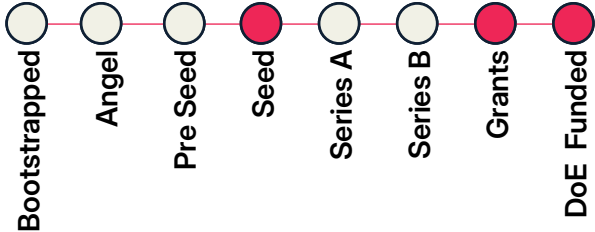
Revenue

2024: ~\$200K

2025: ~\$400K

2026 proj.: ~\$700K

Latest Round of Funding



Funding Raised to Date: \$33 million

Grants, Contracts, Memberships

DoE, DoD, NASA, NSF; First multi-year off-take agreement secured in 2025 with a commercial customer. First joint development agreement signed in 2026 with a Tier 1 global wire and cable manufacturer.; Third Derivative Industrial Innovation Cohort

TAM, Tech, & Supplemental

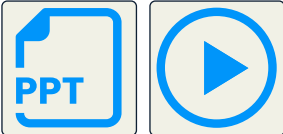
Total Addressable Market

\$230 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

Copper and aluminum are heavy, and weight is a cost every industry pays for: in fuel burned, structures reinforced, cable that's harder to route. Galvorn gives engineers a conductive material that weighs a fifth as much as copper. Ideal customers are aerospace, defense, space, and energy companies designing wire, cable, or shielding where mass matters and a lighter conductor changes what the system can do.

Value Proposition

Galvorn is a new generation of advanced carbon engineered to have it all: metal-level conductivity combined with superior strength, lightweighting, and flexibility in a single material. It cuts system weight without sacrificing durability, resists corrosion and flame for safer, longer-lasting performance, and is fully recyclable at scale — a truly circular, sustainable alternative to copper and aluminum.

Product or Service

DexMat manufactures Galvorn, an advanced carbon material supplied as fiber, wire / yarn, fabric, or film, made through a proprietary wet fiber spinning process at DexMat's Houston, Texas facility. Customers use it in place of copper and aluminum in wire, cable, and shielding applications where weight, flexibility, or fatigue resistance matter, particularly in aerospace, defense, space, and energy.

Founding Team Description

Matteo Pasquali (Co-Founder & Chief Science Advisor) co-invented CNT fiber spinning with Nobel laureate Richard Smalley and spent 20+ years refining it. Dmitri Tsentlovich (Co-Founder & CTO) spent 15 years translating CNT properties into fiber performance. Bryan Guido Hassin (Co-Founder & CEO) has \$400M+ in exits and ran Third Derivative, moving \$2.5B+ into climate tech. Together: the science, manufacturing know-how, and capital experience to scale Galvorn to commercial production and sales.

Technologies & Unique Know-How

DexMat's core technology is a proprietary wet-spinning process that turns carbon nanotubes into continuous, production-scale Galvorn fiber and film, delivering metal-level conductivity plus superior strength, lightweighting, and corrosion/flame resistance in one material. Unlike competitors stuck at lab-scale, DexMat has solved CNT manufacturing scale-up, the barrier that has blocked commercial adoption for two decades, backed by co-founder Pasquali's foundational fiber-spinning IP.

Traction & Competitive Landscape

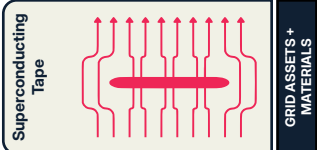
DexMat ships Galvorn today to aerospace, defense, space, and energy customers, and in 2025 signed its first multi-year offtake. In April 2026 it signed a joint development agreement with a Tier 1 global wire and cable manufacturer. Versus copper/aluminum incumbents and lab-scale CNT rivals, DexMat's edge is Galvorn's multi-property advantage plus proven manufacturing scalability.

Pilot Demonstration & Cost

A small-scale pilot would develop and validate grid termination hardware for Galvorn ACNT transmission conductors, fitting connectors and splices to a prototype ACNT line and testing under real mechanical and electrical loads. The ideal partner is a grid termination or cable accessory manufacturer with existing utility relationships and testing infrastructure, whose expertise would accelerate qualification and bring Galvorn ACNT to market faster.

Cost: \$1M

BOREAL CONDUCTORS



Company Overview

Point of Contact Information

Name: Christian Lacroix
Title: CEO
Email: clacroix@borealconductors.com
Phone: (514) 792-0812

Company Information

Legal Name: Boreal Conductors Inc.
Address: 6850 rue de Saint-Vallier, Montreal QC, H2S 2P9
Date Founded: 01/27/2022
Web Address: <https://borealconductors.com/>
Number of Employees: 4FT 1PT

Founding Team

Founders & Leadership

Name: Christian Lacroix
Title: CEO
Email: clacroix@borealconductors.com

Name: Frederic Sirois
Title: CSO
Email: fsirois@borealconductors.com

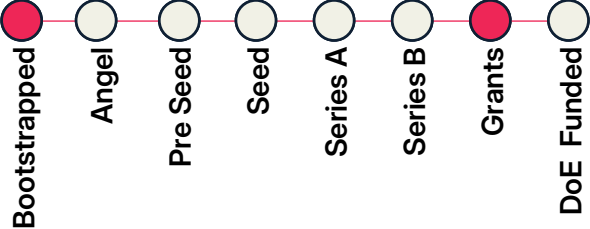


Funding & Traction

Revenue

2024: Confidential
2025: Confidential
2026 proj.: Confidential

Latest Round of Funding



Funding Raised to Date: \$500,000

Grants, Contracts, Memberships

Ministère de l'Économie, de l'Innovation et de l'Énergie (Québec), NRC IRAP; Demonstration of highly resilient superconducting power cables for aircraft with Airbus UpNext - Demonstration of the CFD concept in superconducting tapes for an organization in the fusion industry

TAM, Tech, & Supplemental

Total Addressable Market

\$500 million

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Superconductors are a key element for the construction of tokamaks and stellarators used in nuclear fusion systems to deliver a clean, stable and abundant energy source. Superconductors can be used to bring high power in dense consumption areas such as data centers without any losses. However, superconductors must be protected against quench, which can damage power devices if not well engineered.

Value Proposition

Boreal Conductors offers improved superconducting tape that are resilient against quench and allows to detect much quicker a quench event in superconducting power devices such as superconducting magnets for fusion. Boreal's technology can also be used to fabricate a highly performant superconducting fault current limiter, which is used as a "fuse that doesn't break", to protect power equipment from fault current events and increase the stability and resiliency of the electrical grid.

Product or Service

Boreal Conductors offer a unique improved superconducting tape architecture, called CFD tape, that allows to detect quicker a quench event and to prevent local damages. Using CFD tapes allow to increase the operating current, which reduces the footprint and the cost of the superconducting power device. CFD can be integrated in the form of a cable for power transmission and a coil for magnetic field generation.

Founding Team Description

Co-founders Christian Lacroix and Frederic Sirois are world renowned scientists in applied superconductivity. They are well connected in both the industry and academia. Christian Lacroix is the CEO of Boreal Conductors with a strong background in physics and material science and the main inventor of CFD tapes. He has been growing the business by bringing new customers. Frederic Sirois develops finite-elements modelling of superconductors for customers purposes.

Technologies & Unique Know-How

Boreal Conductors modify superconducting tape by adding a layer called current flow diverter (CFD) in a clever way that enhanced the normal zone propagation velocity (NZPV) upon a quench event. The fabrication method relies on a unique know-how developed over several years. The solution of Boreal Conductors completely reverses the paradigm anchored in the superconductivity industry, opening new possibilities to protect and secure power devices against quench.

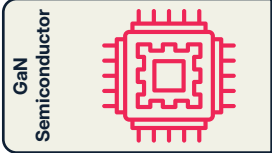
Traction & Competitive Landscape

Boreal Conductors have been working with OEM (contracts) in the fusion industry and aeronautics since its incorporation in 2022. Boreal Conductors is the only manufacturer that offers high NZPV quench resilient superconducting tape. The only other solution is to reduce performance of superconducting tape to ensure adequate quench protection. Boreal Conductors is currently discussing with several potential customers to fabricate demos that includes CFD tapes.

Pilot Demonstration & Cost

Boreal Conductors aim to demonstrate the performance of a superconducting power device based on CFD tapes. It could be a superconducting fault limiter module, a power cable, or a superconducting coil. The ideal partner is a utility that is interested in superconducting technologies to reduce power losses and increases the stability of the grid. Another ideal partner is a superconducting power devices manufacturer that requires better quench protection.
Cost: ~\$100k-\$1M (depends on cryogenics requirements)

Bedrock Semiconductor



Company Overview

Point of Contact Information

Name: Troy Baker
Title: CEO
Email: troy@bedrocksemi.com
Phone: (805) 680-6145

Company Information

Legal Name: Bedrock Semiconductor
Address: Hummelstown, PA 17036

Date Founded: 09/15/2025
Web Address: <https://bedrocksemi.com/>
Number of Employees: 2FT 2PT

Founding Team

Founders & Leadership

Name: Troy Baker
Title: CEO
Email: troy@bedrocksemi.com

Name: Max Shatalov
Title: CTO
Email: max@bedrocksemi.com

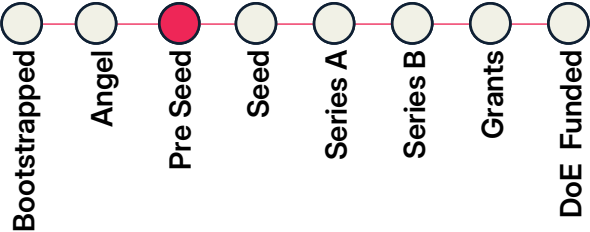


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: N/A

Latest Round of Funding



Funding Raised to Date: \$100,000

Grants, Contracts, Memberships

AlphaLab (Pittsburgh), Plug and Play

TAM, Tech, & Supplemental

Total Addressable Market

\$25 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

Data centers represent a large demand on electricity, yet some of the electricity they consume is wasted as inefficient heat losses when converting voltage. High-voltage GaN power devices will save energy in data centers by their low losses compared to incumbent solutions when incorporated into solid-state transformers. Customers include electrical equipment vendors such as ABB, Schneider Electric, Eaton, and Vertiv, and Joules Accelerator alumni DG Matrix and Ionate.

Value Proposition

Bedrock Semiconductor develops high-performance GaN RF and high-voltage power devices using native GaN wafers. This approach results in the highest efficiency of both RF and power devices when compared to other GaN devices and compared to all other commercially viable semiconductor technology. The incumbent technology for the high-voltage power device market is based on SiC, however GaN has 5X lower resistance and 10X faster switching speed due to the fundamental materials properties.

Product or Service

The company plans to sell high voltage GaN power devices, competing with SiC power devices. In addition, the company plans to sell high performance GaN RF devices.

Founding Team Description

Troy Baker, PhD, (CEO): Serial Entrepreneur and previously founder of a company in China that produces 4" GaN wafers.
Max Shatalov, PhD (CTO): Expert in GaN technology encompassing materials development, devices, equipment, and IP.
Dirk Ehrentaut, PhD (VP Materials): Expert on GaN wafers, having worked on novel GaN growth technology.
Travis Anderson, PhD (VP Devices): Expert on GaN devices, currently professor at the University of Florida and formerly at the U.S. NRL.

Technologies & Unique Know-How

The key enabling technology for GaN-on-GaN devices is the GaN substrate wafer. Yet only 3 companies around the world can produce 4" GaN wafers; 1 in China and 2 in Japan. The founder of Bedrock, Troy Baker, is also the founder of the Chinese company that produces 4" GaN wafers. In addition, the team is well rounded in expertise including GaN wafer production, GaN devices, and GaN growth equipment.

Traction & Competitive Landscape

The company is currently raising \$10M in order to build GaN production equipment. We have completed an NSF I-Corps study and continue to verify customer interest in the products.

There are 4 companies in the US working on GaN-on-GaN devices; 2 public companies and 2 startups. None of them have GaN wafer technology or the enabling GaN growth equipment technology

Pilot Demonstration & Cost

We would like to test our GaN power devices in solid-state transformers and similar equipment. ABB, DG Matrix, and Ionate would be ideal partners.

Cost: We are raising \$10M seed round in order to establish an R&D lab in Pittsburgh and demonstrate 4" GaN wafers and prototype GaN devices.

SONDER LABS

Sonder Labs

Power Balancing



GRID ASSETS + MATERIALS

Company Overview

Point of Contact Information

Name: Marcus Svensson Dahlin
Title: Senior Director Partnerships
Email: marcus@sonderlabs.com
Phone: +46702070923

Company Information

Legal Name: Sonder Labs
Address: Banvaktsvägen 20, 171 48 Solna, Sweden
Date Founded: 12/09/2024
Web Address: <https://sonderlabs.com/>
Number of Employees: 7FT 2PT

Founding Team

Founders & Leadership

Name: Andreas Haas
Title: CEO
Email: andreas@sonderlabs.com



Name: Peter Carlsson
Title: Chief Growth Officer & Chairman
Email: peter@sonderlabs.com

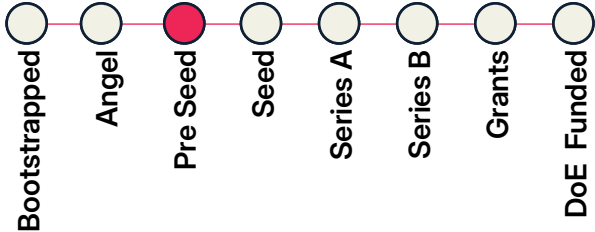


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: N/A

Latest Round of Funding



Funding Raised to Date: \$5.1 million

Grants, Contracts, Memberships

Swedish Energy Agency for "verification of concept design" grant of SEK500 000 (~USD 53 000)

Letter of support from large industrial integrator of grid solutions such as E-STATCOMs.

TAM, Tech, & Supplemental

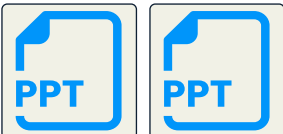
Total Addressable Market

\$125 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

On the generation side, grids lose inertia as renewables replace synchronous generators, while on the user side AI factories and electrified industries add fast, high-magnitude load swings. Both create a need for high-power transient balancing over 10-90 seconds, which today's supercaps and batteries cannot serve. Sonder's neocapacitor closes this gap. Ideal customers are power quality integrators (Hitachi Energy, S&C Electric, GE Vernova, etc.) who deploy solutions across applications/markets.

Value Proposition

Sonder's neocapacitor is a hybrid capacitor that brings a step change in power balancing technology. Based on development, prototyping and modelling to date it delivers 4-5 times the continuous power of a supercapacitor and 2.5 times higher energy density, while maintaining similar cycle life (millions of cycles). This unlocks efficient transient balancing over 10-90 seconds at far smaller footprint, complexity and cost. Built entirely from abundant, non-critical materials, it is globally scalable.

Product or Service

Sonder sells neocapacitor racks, not bare cells. Each rack is finished hardware that integrates into a customer's electrical platform alongside standard power electronics, so integrators and utilities can deploy it without redesigning their systems. One rack product serves two markets. On the grid it provides virtual inertia, in data centers it handles compute rack balancing. Selling at rack level lets Sonder stand behind full-unit performance while fitting infrastructure customers already run.

Founding Team Description

Sonder's founders have scaled electrochemical hardware from lab to mass production before. CEO Andreas Haas led next-gen cell development at Northvolt, taking new chemistries from R&D to market. Chairman Peter Carlsson founded Northvolt and was VP Supply Chain at Tesla. Around them sits a team that includes the core inventor of the neocapacitor technology and seasoned electrochemical experts, with cell development and manufacturing experience from Natron, Tesla and Panasonic.

Technologies & Unique Know-How

Sonder's neocapacitor is the first hybrid capacitor to pair a Prussian Blue Analogue cathode with a capacitive carbon anode, a new class closing the gap between supercaps and batteries. Where supercaps store charge only electrostatically, Sonder adds fast faradaic intercalation into the cathode's open 3D framework, unlocking far higher continuous power. Sonder has filed first patents here, builds on abundant non-critical materials, and the team has long-standing experience within the chemistry.

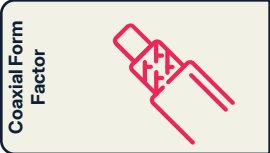
Traction & Competitive Landscape

Sonder is pre-commercial. The cell, our core innovation, is at TRL5 in prototype, with first module prototypes due Q3 2026. Active development partnerships with European power quality integrators contribute test infrastructure and validation. The market splits between supercaps (Skeleton, Maxwell), too low in energy to sustain long transients, and costly niobium super-batteries (Nyobolt) reliant on concentrated supply. Sonder's higher continuous power fills the 10-90 second gap between them.

Pilot Demonstration & Cost

Following module and rack validation, a pilot would package Sonder racks into a productized 20ft container, integrated alongside power electronics and validated under representative grid duty cycles against a supercapacitor baseline. Ideal partners, and the introductions Sonder would most value, are transformer and switchgear providers building the electrical platform (EPC Power, Dynapower, GE Vernova, S&C Electric) and US utilities such as Duke Energy supplying real grid load profiles. Cost: \$1.5 million

SCALED POWER SOLUTIONS



GRID ASSETS + MATERIALS

Company Overview

Point of Contact Information

Name: Mark Cairnie

Title: Founder

Email: mcairnie@scaledpowersolutions.com

Phone: (484) 709-5164

Company Information

Legal Name: Scaled Power Solutions, LLC

Address: Arlington, VA 22204

Date Founded: 06/12/2025

Web Address: <https://www.scaledpowersolutions.com/>

Number of Employees: 1FT

Founding Team

Founders & Leadership

Name: Mark Cairnie

Title: Founder

Email: mcairnie@scaledpowersolutions.com

Name:

Title:

Email:

Funding & Traction

Revenue

2024: N/A

2025: N/A

2026 proj.: N/A

Latest Round of Funding

Bootstrapped

Angel

Pre Seed

Seed

Series A

Series B

Grants

DoE Funded

Funding Raised to Date: \$300,000

Grants, Contracts, Memberships

Advanced Research Project Agency - Energy (ARPA-E), Dominion Energy Innovation Center (DEIC), Virginia Tech LAUNCH Office, Virginia Clean Energy Innovation Bank (VCEIB); The company is a subawardee on ARPA-E award, responsible for leading T2M and commercialization milestones, while the Virginia Tech (program PI) leads the technical milestones

TAM, Tech, & Supplemental

Total Addressable Market

\$1.2 billion

Tech Readiness Level

1

2

3

4

5

6

7

8

9

Supplemental Documents

Company Description

Problem Statement & Target Customer

AI-driven datacenter load growth is outpacing lead times for conventional medium-voltage equipment. Utilities face the same bottleneck from lengthening interconnection queues and space-constrained sites. Scaled Power closes this gap with a cable-integrated coaxial platform delivering medium-voltage power conversion in a fraction of the footprint, cost, and time. Ideal customers: datacenter operators racing to bring capacity online, and utilities managing space-constrained upgrades.

Value Proposition

We sell a form factor advantage, not a functional one: same medium-voltage conversion, a fraction of the footprint, faster, simpler install. GIS switchgear already proves footprint commands a 2x premium; we capture comparable or greater value. Sources of value: avoided land cost, avoided delay cost, avoided capital-timing risk for utilities, and avoided outage cost from easier undergrounding. One base cell serves datacenter, utility, industrial, and defense segments.

Product or Service

A modular, coaxial power electronics platform sold as direct capital equipment, not a subscription or service. A common base cell configures into products spanning datacenter rack-level power delivery to full substation-scale units, integrating medium-voltage conversion into a compact, cable-based form factor.

Founding Team Description

Founder Mark Cairnie holds a Ph.D. from Virginia Tech's Center for Power Electronics Systems and is a named co-inventor on four of five patents underlying the platform, giving direct technical grounding from the technology's origin. A Power Electronics Engineering Lead and Business Development Lead are planned as first hires post-close.

Technologies & Unique Know-How

Five pending patents span coaxial semiconductor packaging, converter architecture, topology, and magnetics/capacitor design, protecting multiple structural layers, not one circuit trick, making the approach hard to design around. The IP originated from a multi-institution research program (Virginia Tech, UConn, National Laboratory of the Rockies) with prior ARPA-E validation.

Traction & Competitive Landscape

Pre-pilot stage: closing a pre-seed round to fund pilot planning, facility buildout, and pilot LOIs across datacenter and utility customers, alongside a parallel ARPA-E SCALEUP application. Competitors include conventional switchgear/transformer incumbents, GIS vendors, modular substation makers, and emerging solid-state entrants. Our cable-integrated architecture combines the smallest footprint and fastest deployment across all categories.

Pilot Demonstration & Cost

Our first pilot is a small utility sandbox demonstrator, a single/dual-cell unit stepping 4,160V down to 480V at a modest power level, tested in a controlled grid-simulator environment before any field deployment. Ideal partner: a utility willing to host bench-scale validation in a non-production sandbox setting. Estimated cost: \$300K-\$600K for hardware, assembly, and testing, leveraging the core team and facility funded by pre-seed.

Company Overview

Point of Contact Information

Name: Thomas Cowley
Title: CEO
Email: thomas.cowley@runpilot.io
Phone: 07468418249

Company Information

Legal Name: AENI Limited (dba RunPilot)
Address: The Innovation Centre, 217 Portobello, Sheffield, S1 4DP
Date Founded: 4/11/2026
Web Address: <https://www.runpilot.io/>
Number of Employees: 2FT 5PT

Founding Team

Founders & Leadership

Name: Thomas Cowley
Title: CEO
Email: thomas.cowley@runpilot.io



Name: Joseph Hammond
Title: CTO
Email: joseph.hammond@runpilot.io

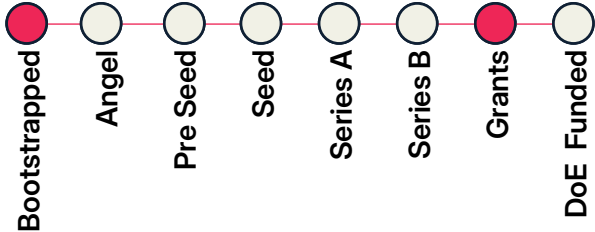


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$150,000

Latest Round of Funding



Funding Raised to Date:

Grants, Contracts, Memberships

University of Sheffield, Pilot with Tier-1 engineering consultancy (under NDA); Subject-to-contract sub-contracted to Atkins for an innovation project with partners including National Grid, SSEN, NESO, Regional contract; Accepted onto the National Climate Tech Accelerator, delivered by Sustainable Ventures

TAM, Tech, & Supplemental

Total Addressable Market

\$3.4 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

The US needs faster grid and utility buildout, but planning workflows are being asked to move beyond what current processes allow. Route, land, engineering, cost, permitting and stakeholder evidence sits across separate teams and tools. When assumptions change, teams revisit options, rerun analysis and rebuild evidence. This slows decisions and lets higher risk options progress before delivery risks are clear. Target customers are utilities, developers, cooperatives and planning consultants.

Value Proposition

RunPilot helps planning teams make faster, better evidenced infrastructure decisions before options become expensive to change. It brings cost, land, permitting, environmental, engineering, stakeholder and objection exposure into one comparison workflow, so teams can generate, test and compare route, connection, substation, network and phasing options as assumptions change.

Product or Service

RunPilot is a software platform for infrastructure planning teams. We sell configured project workspaces and team or enterprise licences to utilities, developers and consultants. The RunPilot planning engine converts project, public and customer supplied data into cost, constraint and exposure surfaces, then produces comparable options, maps, risk outputs, assumption logs and evidence reports.

Founding Team Description

RunPilot was created by Dr Thomas Cowley and Joseph Hammond during their PhDs in infrastructure routeing, energy systems, spatial analysis, optimisation, socio technical modelling and engineering software. Professor Solomon Brown adds over 20 years of energy systems and industry research expertise. The team combines deep modelling knowledge with practical product development.

Technologies & Unique Know-How

RunPilot's core technology is a planning engine combining geospatial processing, optimisation, AI assisted spatial analysis and infrastructure specific engineering logic. It generates and compares route, connection, substation, network and phasing options, with parcel exposure, stakeholder exposure, scenario testing and assumption tracking to assess how delivery risk changes as evidence matures.

Traction & Competitive Landscape

RunPilot has built and tested its core platform on real infrastructure planning use cases, including UK industrial cluster work across Humber and Teesside and benchmarking against published UK National Grid routeing material. Commercial traction includes a scoped paid evaluation under NDA, a subject to contract subcontract on an Arup led Strategic Innovation Fund project and wider utility planning pipeline.

Pilot Demonstration & Cost

A small scale pilot would focus on one live or representative grid planning problem, such as a grid upgrade, interconnection, substation siting issue or route comparison. RunPilot would ingest available project, public and customer supplied data, configure assumptions and generate comparable options. Success would be clearer option comparison, earlier risk visibility and reduced manual rework.
Cost: \$50,000 - \$100,000 depending on complexity



Company Overview

Point of Contact Information

Name: Mohak Mangal
Title: Founder & CEO
Email: mohak.mangal@pravah.com
Phone:

Company Information

Legal Name: Pravāh
Address: 625 2nd Street, Suite 200, SF, CA, 94107
Date Founded: 10/17/2025
Web Address: <https://www.pravah.com/>
Number of Employees: 15 FT 5PT

Founding Team

Founders & Leadership

Name: Mohak Mangal
Title: CEO
Email: mohak.mangal@pravah.com
Name: Dhruv Suri
Title: CTO
Email: dhruv.suri@pravah.com

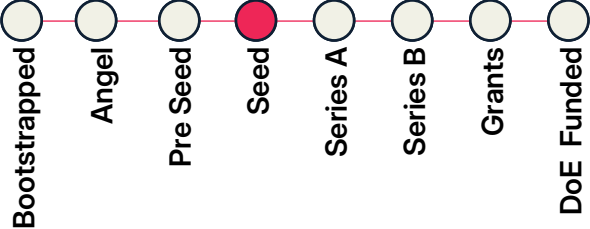


Funding & Traction

Revenue

2024: N/A
2025: \$1,500,000
2026 proj.: \$15,000,000

Latest Round of Funding



Funding Raised to Date:

Grants, Contracts, Memberships

\$1.5M contract with one of India's largest utilities to build a granular GIS map of their grid using satellite and street-view imagery, part of our broader work with 10+ utilities across the US, India, and Brazil, including EPB and ERCOT. Google AI, NVIDIA, Flexlab, Powerthon AI, PearX, Embed

TAM, Tech, & Supplemental

Total Addressable Market

\$40 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

US utilities face two compounding problems: weather-driven outages costing the US economy ~\$150B/year (this weekend's storms left 800,000+ customers without power), and interconnection studies too slow to keep up with the data center buildout, taking years with cost estimates swinging by hundreds of percent. Our ideal customer is a US investor-owned or municipal utility running a grid-hardening program, or an RTO like ERCOT facing a growing large-load interconnection queue.

Value Proposition

We turn imagery and grid data into faster, defensible decisions for utilities and RTOs. On distribution, we target hardening capital precisely using satellite/street-view imagery and outage history instead of manual review. On transmission, our ML model gives power flow solvers a smart starting point, cutting interconnection study time from hours to seconds.

Product or Service

Two ML products sold as software plus engineering partnership: (1) a grid-hardening and outage-prioritization engine that screens the network from satellite/street-view imagery and ranks spans to harden first; (2) a machine-learning power flow model that warm-starts conventional solvers, letting utilities and RTOs clear far more interconnection studies through the same queue.

Founding Team Description

Mohak Mangal (Stanford MBA, ex-World Bank economist), Dhruv Suri (Stanford PhD in Energy, built grid ML at Google X, published on peaker plants and data center buildout), and Shourya Bose (PhD UC Santa Cruz, Argonne National Lab, ML for power flow and dispatch). Rounded out by Aman Gupta (co-developed NASA/IBM's Prithvi WxC) and Helgi Hilmarsson (Stanford MS, graph neural networks).

Technologies & Unique Know-How

Graph neural networks trained on real grid topology solve power flow in one pass instead of iterating from scratch. Computer vision over satellite and street-view imagery scores pole and vegetation risk across thousands of miles automatically. Physics-based load flow models prioritize capex spend for reliability and affordability. Our team has published this research and deployed it on real utility networks.

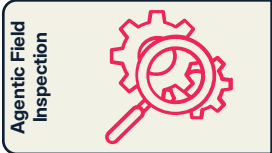
Traction & Competitive Landscape

We work with 10+ utilities and grid operators across the US, India, and Brazil: EPB (hardening), ERCOT (interconnection), PG&E, MSEDCL, UPPCL, APEPDCL, TPDDL, and Energisa. Alternatives, Itron, Oracle Utilities, GE, and manual GIS/engineering consultancies, are slow, manual, or single-purpose. We are the only platform combining imagery, power flow, and capex prioritization into one system.

Pilot Demonstration & Cost

A focused 90-day sprint on a utility's own data: the utility shares historical outage records, network shapefiles, and imagery access for one part of its system, and we independently produce a ranked, image-supported hardening or interconnection-study recommendation, benchmarked against what actually happened. Ideal partner: a US utility with an active hardening program, or an RTO with a growing interconnection queue.
Cost: \$50,000 - \$100,000

CLONEABLE AI



Company Overview

Point of Contact Information

Name: Patrick Lohman
Title: Co-Founder / CRO
Email: pat@cloneable.ai
Phone: (260) 312-5640

Company Information

Legal Name: Cloneable AI
Address: 223 S. West St, Ste. 900, Raleigh, NC, 27603
Date Founded: 3/30/2023
Web Address: <https://www.cloneable.ai/>
Number of Employees: 10FT 2PT

Founding Team

Founders & Leadership

Name: Patrick Lohman
Title: CRO
Email: pat@cloneable.ai



Name: Lia Reich
Title: CEO
Email: lia@cloneable.ai

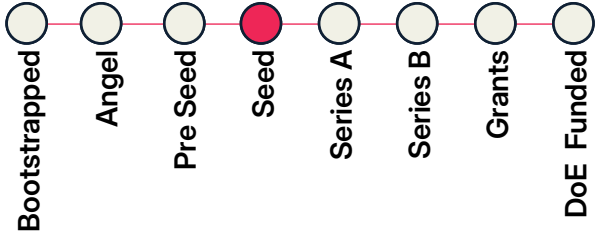


Funding & Traction

Revenue

2024: Confidential
2025: Confidential
2026 proj.: Confidential

Latest Round of Funding



Funding Raised to Date: \$5,350,000

Grants, Contracts, Memberships

Notable utility customers- AEP & SoCal Edison. Notable Utility EPC customers- TRC, Burns & McDonnell

TAM, Tech, & Supplemental

Total Addressable Market

\$68 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Utilities and utility engineering firms face two bottlenecks: field crews rarely capture what the back office needs to make decisions, and the engineering work that follows (pole loading, make-ready, permitting) is manual and locked inside tools like SPIDAcac and ESRI that don't talk to each other. Our ideal customer is a utility engineering firm or IOU that is winning more work than their current headcount can process.

Value Proposition

Cloneable connects two workflows no single platform has solved together: an AI-guided iPhone app that captures accurate pole data in the field, and a desktop AI agent that automates the engineering work that follows inside the software teams already use. The result is 30 to 100x throughput on tasks like pole loading and make-ready, without replacing existing systems.

Product or Service

Cloneable sells two connected software products: Cloneable Field, an iOS app that guides field crews through AI-verified pole data collection, sold as a SaaS license or embedded SDK. Cloneable Agent automates back-office engineering workflows inside software like SPIDAcac, O-Calc, or ESRI, and is sold on a usage-based model tied to workflow volume.

Founding Team Description

All three co-founders were among the first employees at Precision-Hawk, where utility inspection became the dominant use case and the team worked with the top 5 of 10 IOUs in the US. Pat Lohman (CRO) leads sales and customer relationships. Lia Reich (CEO) drives company strategy and execution. Tyler Collins (CTO) leads product strategy and engineering. The team has built together for 15 years across two companies with deep roots in the utility software & field services ecosystem.

Technologies & Unique Know-How

Cloneable's core technologies are an AI-guided mobile data capture engine built for utility field conditions, and a desktop AI agent that learns company-specific workflows and operates inside existing engineering software without API integrations. Rather than rebuilding integrations for every stack, the agent bridges systems the way a trained engineer would. Fifteen years inside utility workflows gives our team fluency in the nomenclature and tooling no general-purpose AI platform can replicate.

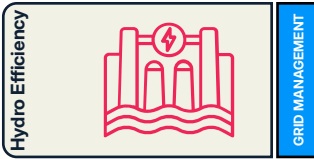
Traction & Competitive Landscape

Cloneable has ~50 customers including Burns & McDonnell, TRC, AEP, and SCE, and recently closed a \$4.5M seed round. Competitors address field collection or back-office automation in isolation. No platform connects both within the utility engineering workflow. General-purpose AI tools lack the domain fluency; niche engineering software vendors lack the AI layer. Cloneable owns the full workflow.

Pilot Demonstration & Cost

An ideal pilot runs 60 to 90 days with a utility or engineering firm that has an active pole inspection or make-ready workflow. For Cloneable Field, a field crew of 2 to 5 collects data on a defined pole set, measured against their current process. For Cloneable Agent, we train on one back-office workflow in pole-loading, design or GIS and measure throughput against a human baseline. Ideal partners include IOUs, engineering firms, or software vendors looking to embed AI into their platform. Cost: \$25,000 - \$75,000

AXOLOTIS.AI



Company Overview

Point of Contact Information

Name: Wilkins Aquino
Title: CEO
Email: wilkins.aquino@axolotisopt.com
Phone: (607) 262-0756

Company Information

Legal Name: Axolotis.ai, Inc.
Address: 7411 Montibillo pkwy, Durham, NC 27713
Date Founded: 5/1/2025
Web Address: <https://www.axolotisopt.com>
Number of Employees: 3FT 2PT

Founding Team

Founders & Leadership

Name: Wilkins Aquino
Title: CEO
Email: wilkins.aquino@axolotisopt.com

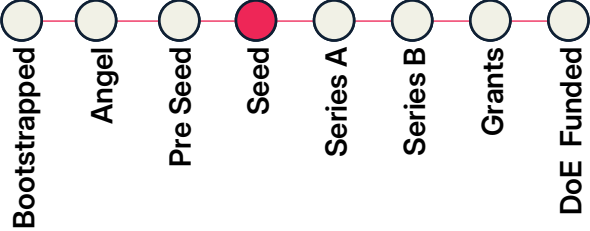
Name:
Title:
Email:

Funding & Traction

Revenue

2024:
2025:
2026 proj.: \$100,000

Latest Round of Funding



Funding Raised to Date: \$2,250,000

Grants, Contracts, Memberships

TAM, Tech, & Supplemental

Total Addressable Market

\$3.4 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

We aim at ensuring that every drop of energy fuel is used as efficiently as possible—turning power plants into a smart, flexible, and data-driven cornerstone of the global clean energy transition, with human expertise guiding every critical decision. Ideal customers are utilities and independent power producers.

Value Proposition

We discover hidden value that is left on the table in daily operations. That is, we unlock more power from existing assets through higher efficiency, reliability, and sustainability. We can increase net effective capacity with a minimal investment and no new infrastructure.

Product or Service

We offer a proprietary optimization and agentic AI framework that: Maximizes energy produced per unit of fuel, reduces inefficiencies such as spillage and under-generation, enhances operational compliance and environmental stewardship, and Integrates seamlessly with storage and market operations for dispatch flexibility. Empowers operators with AI-driven insights while maintaining human oversight of critical decisions. Ideal customers are utilities and independent power producers.

Founding Team Description

We have a deep and unique synergy in hydroelectric plant operations and optimization experience. Our team of leaders include: Dr. Wilkins Aquino—Founder & CEO, Axolotis; 20+ years in optimization, ML, and scientific computing consultant to Sandia National Laboratories. Dr. Kristina Johnson – Senior Advisor; Founder of Cube Hydro; Catalyzer Ventures, former U.S. Under Secretary of Energy. Dr. Neal Simmons-Senior Advisor; CEO of Eagle Creek Renewable Energy

Technologies & Unique Know-How

We possess a holistic framework that integrates efficient nonlinear optimization solvers, machine learning, and agentic AI.

Traction & Competitive Landscape

We are in final negotiations for our first client contract with an independent power producer. In addition, we have completed one pilot, are in the middle of three more, and are waiting on data for a fourth one. We differentiate from our competitors by adopting a physics-based optimization that embraces the intricacies of hydro systems including, but not limited to, cascading effects, integer constraints, nonlinear efficiency relations, automatic generation control, and storage.

Pilot Demonstration & Cost

In a pilot, we use historical data (hourly generation, water levels, flow, operational and environmental constraints, etc.) and determine optimal generation schedules that are then compared to as-run conditions. We then compute potential gains if our solutions are implemented. This process takes from 1 to 12 weeks depending on the complexity of the plants. Ideally, we would work with utilities and IPPs.
Cost: \$10,000 - \$25,000

Company Overview

Point of Contact Information

Name: Anna Bazley
Title: Head of Government and Regulatory
Email: anna.bazley@enodatech.com
Phone: +447913271066

Company Information

Legal Name: Enoda Ltd.
Address: Quartermile 3, 10 Nightingale Way, Edinburgh, Scotland, EH3 9EG
Date Founded: 7/28/2021
Web Address: <https://www.enoda.com/>
Number of Employees: FT 134 PT 7

Founding Team

Founders & Leadership

Name: Andrew Scobie
Title: CTO & CPO
Email: ascobie@enodatech.com



Name: Paul Domjan
Title: Chief Policy & GAO
Email: pdomjan@enodatech.com

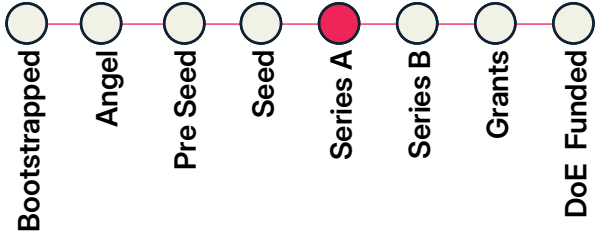


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$500,000

Latest Round of Funding



Funding Raised to Date: ~\$30 million

Grants, Contracts, Memberships

Operational pilot agreements with Spanish DSO (UFD-Naturgy) and major Polish DSO (currently confidential). MoUs in place with a Polish DSO within a top-three European utility group and US publicly-owned utility.

TAM, Tech, & Supplemental

Total Addressable Market

\$160 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

As the share of switch loads (e.g. digital loads) and inverter-based resources increases, distribution networks face worsening voltage and frequency instability, power quality and congestion, limiting their ability to serve load growth. Enoda's solution combines power-electronics hardware and software to shape the AC signal, regulate power flow, and coordinate distributed assets in real time. Its target customers are electric utilities seeking more feeder capacity, resilience and control.

Value Proposition

Enoda transforms distribution substations from passive grid bottlenecks into actively controlled nodes. Its PRIME Exchanger combines network regulation capabilities in one drop-in platform, replacing functions otherwise requiring multiple devices. Combining with proprietary software, Enoda gives utilities real-time visibility, secure distributed control and fleet-level balancing capability, enabling more DERs, resilience and network capacity without relying solely on traditional reinforcement.

Product or Service

ENODA sells an integrated grid-modernization platform for electricity distribution networks. Its PRIME Exchanger is a drop-in substation device that combines voltage regulation, power factor control, harmonic suppression, phase balancing, bidirectional power flow and network data in one system. The edge-to-cloud Nodal Asset Management and ENSEMBLE platforms gives utilities real-time visibility, secure control, balancing capability and more capacity for DERs and renewables in the local network.

Founding Team Description

"Enoda's founding team combines depth across invention, policy, operations, and commercial. Andrew Scobie is an inventor and systems designer with energy innovation patents and experience leading advanced engineering firms; Paul Domjan is a former CEO and senior energy security adviser with deep policy and market knowledge; and Jacqui Porch adds energy-sector branding, product commercialization and capital-raising experience.

Technologies & Unique Know-How

ENODA's core technology is dynamic flux modulation. The PRIME Exchanger uses primary, secondary and control coils, advanced control algorithms, and power electronics to calculate and inject phase-specific flux correction signals in the torus core, shaping each output waveform in real time. ENODA is positioned to lead because no other single device combines these functions dynamically, and its NAM and ENSEMBLE stack turns novel hardware into a deployable, utility-ready platform for use today.

Traction & Competitive Landscape

ENODA is currently ahead of a still fragmented field of smart-transformer and grid-edge players. Competitors are focused on narrower add-on or earlier-stage solutions, whereas ENODA's hardware plus software platform offering is differentiated by its breadth; combining multiple grid functions dynamically in one device, with NAM and ENSEMBLE adding control, data and balancing value. ENODA has live grid trial activity in two countries with DSO partners Naturgy (Spain) and a major Polish DSO.

Pilot Demonstration & Cost

A small-scale pilot would place one or a few PRIME Exchangers at a utility distribution substation or constrained feeder to test voltage regulation, phase balancing, harmonics, power factor control and visibility through NAM under live conditions. The ideal partner is a forward-leaning electric utility with DER growth, power quality issues, and a live feeder where performance can be measured against baseline operations. Cost: \$250k-1m, depending on number of units, site integration requirements, commissioning scope and monitoring.

WATCHFULY



Asset Health + Power Flow

GRID MANAGEMENT

Company Overview

Point of Contact Information

Name: Nikhil Kriplani
Title: CEO
Email: nikhil.kriplani@watchfully.com
Phone: (919) 302-7457

Company Information

Legal Name: Watchfully Inc.
Address: 505 Commons Walk Cir, Cary NC 27519
Date Founded: 02/05/2025
Web Address: <https://www.watchfully.com/>
Number of Employees: 2FT 2PT

Founding Team

Founders & Leadership

Name: Nikhil Kriplani
Title: CEO
Email: nikhil.kriplani@watchfully.com

Name: Andy Graven
Title: COO
Email: andy.graven@watchfully.com

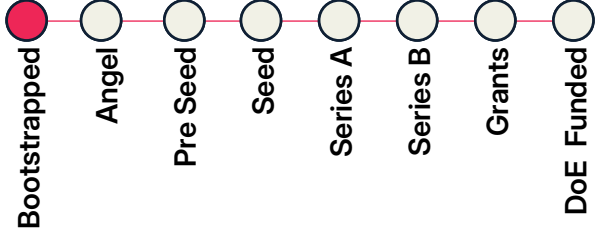


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$50,000

Latest Round of Funding



Funding Raised to Date:

Grants, Contracts, Memberships

TAM, Tech, & Supplemental

Total Addressable Market

\$5 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

Distribution workflows rely on nameplate ratings and periodic study artifacts — stale the moment filed — and asset inspection data never connects to planning or operations. This results in subpar capital prioritization, lost interconnection revenue, and undetected overloads. What's missing is a live control center view of the distribution grid grounded in actual asset condition. The ideal customer is a Grid Reliability VP accountable for uptime, new load connections, and capital efficiency.

Value Proposition

The Plexus product is the only platform that does both AI-powered asset inspection and live agentic power flow — closing the gap between what utilities see in the field and what their grid models reflect. The result is a live control center view grounded in actual asset condition: capital ranked by outage probability reduction per dollar, interconnection headroom certified without manual study setup, and dispatch optimized against true constraints.

Product or Service

Watchfully offers a tiered SaaS subscription. Tier 1: inspection intelligence — condition-accurate asset health scores from existing asset imagery, no new flights or labeled data required. Tier 2: live grid intelligence — a continuously updating power flow engine that tells utilities which assets to replace first, where new load connections are safe to approve, and how to dispatch batteries against true grid constraints. Fixed per-asset and network size-based pricing.

Founding Team Description

The team uniquely blends power systems, defense-grade AI, and enterprise operations. Nikhil (CEO) is a PhD EE who built multi-physics circuit simulators from first principles and deployed AI to DoD programs of record as defense tech CTO. Matt (AI) deploys foundation models for the intelligence community. Alex (Product) former FPL data analyst and Dept. of Navy product deployments. Andy (COO) scaled operations ranging from \$5M to \$350M. Advisor: Davis Montenegro Martinez, EPRI OpenDSS co-author.

Technologies & Unique Know-How

Plexus introduces live agentic power flow to distribution operations — a capability no ADMS or inspection platform has built. Two novel foundations make it possible: asset health-to-power flow mappings, which replaces nameplate assumptions with AI-inspection-grounded circuit parameters, and a custom self-supervised vision model generating asset health scores from existing imagery with zero labeled examples. The team brings DoD foundation model experience and OpenDSS authorship.

Traction & Competitive Landscape

FPL pilot under internal business case review. SDG&E pilot scoping active. ARCOS/AEP co-sell structured. NRECA facilitating co-op introductions. DOE Section 40101 TA engagement active with NLR. EPRI OPAI Consortium member. Esri startup partner. Brattle Group (2026) validates the latent grid capacity thesis. Competitively, ADMS platforms run static one-way models and manual workflows and inspection AI vendors produce flags not agentic power flow grid awareness. Plexus is a category of one.

Pilot Demonstration & Cost

Meets the utility where they are. GIS topology alone is enough to start — Plexus runs live power flow and identifies load connection headroom. Inspection imagery unlocks asset health derating. AMI or SCADA enriches outputs further. Deliverables at any data tier: which assets to replace first, where new load connections are safe, and how to dispatch batteries against true constraints. Ideal partner: an IOU, co-op, or muni with a DER backlog, capital planning pressure, or O&M optimization goals. Cost: \$30,000

CUBENEXUS



Company Overview

Point of Contact Information

Name: Steven Brandt
Title: CEO
Email: steven@cubenexus.xyz
Phone: 9372035630

Company Information

Legal Name: CubeNexus
Address: Chicago, IL, 60607

Date Founded: 4/19/2024
Web Address: <https://cubenexus.xyz/>
Number of Employees: 4 FT 1 PT

Founding Team

Founders & Leadership

Name: Steven Brandt
Title: CEO
Email: steven@cubenexus.xyz

Name: Shilpa Tiwari
Title: CPO
Email: shilpa.tiwari@cubenexus.xyz

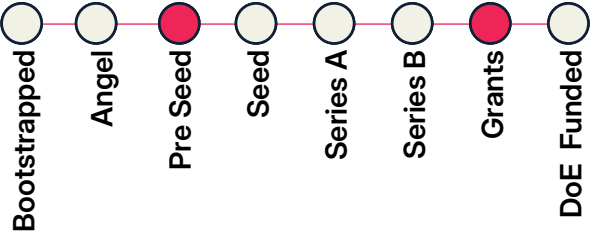


Funding & Traction

Revenue

2024: 0
2025: \$20,000
2026 proj.: \$800,000

Latest Round of Funding



Funding Raised to Date: \$805,000

Grants, Contracts, Memberships

Build in Tulsa

TAM, Tech, & Supplemental

Total Addressable Market

\$850 Million

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Enterprises struggle to turn fragmented, multi-modal spatial data into timely, accurate decisions across energy and industrial systems. CubeNexus solves this by unifying data across space and time with AI-ready precision. Our ideal customers are industrial and energy operators and, more often, software and data platform providers that need a scalable spatial data layer to improve data usability, time-to-insight, and decision accuracy.

Value Proposition

CubeNexus enables enterprises to turn fragmented, multi-modal spatial data into trusted, actionable intelligence. Our cube-based spatiotemporal data layer and AI agents unify data across domains, preserve precision, and dramatically improve data usability, time-to-insight, and decision accuracy—delivered as SaaS or embedded licensing for scalable enterprise deployment.

Product or Service

CubeNexus is an enterprise AI and data infrastructure software solution that unifies multi-modal, multi-domain spatial data into a scalable 4D (space-time) model. Delivered as AI agents and a licensable SDK, CubeNexus enables faster, more accurate analytics and decision-making across energy, industrial, and infrastructure systems by improving data usability, precision, and time-to-insight.

Founding Team Description

The CubeNexus founding team combines deep enterprise AI product leadership with proven experience commercializing technology in regulated, infrastructure-scale markets. Our CPO has 20+ years building and scaling AI and data platforms at Microsoft, Amazon, and startups, taking products from 0 to 1 to production. Our CEO brings extensive experience navigating enterprise and government procurement, partnerships, and go-to-market across defense, energy, and critical infrastructure.

Technologies & Unique Know-How

CubeNexus is built on a proprietary cube-based spatiotemporal data model (TULSA) protected by multiple patents and trade secrets. Our technology unifies multi-modal, multi-domain data into an AI-ready 4D structure that bounds uncertainty, preserves precision, and improves time-to-insight. This foundation enables scalable AI agents and embedded licensing, positioning CubeNexus as a core data layer for industrial and energy platforms.

Traction & Competitive Landscape

CubeNexus has early traction with three paid design partners across energy, aviation, and AI data applications, plus completed pilots validating our technology across sub-surface to air domains. We compete with ESRI, Palantir, C3.ai, Cesium, and Heavy.ai, which focus on maps or data workflows. CubeNexus differentiates with agentic AI-native architecture, global space-to-sub-surface 4D modeling, true time-and-space accuracy, and more cost-efficient, decision-ready insights.

Pilot Demonstration & Cost

A CubeNexus pilot is a small-scale deployment with a utility or grid-adjacent partner to unify multi-modal data (e.g., SCADA, sensors, geospatial, time-series) into a single 4D operational view. The pilot includes a design deep dive, data mapping, and deployment of real-time and historical analytics to improve situational awareness, time-to-insight, and decision accuracy without replacing existing systems.
Pilot Cost: \$95,000

HOLDPOINT



Energization Readiness

GRID MANAGEMENT

Company Overview

Point of Contact Information

Name: Cameron Harris

Title: Founder & CEO

Email: cameron@holdpoint.com

Phone: +61 434 790 083

Company Information

Legal Name: Holdpoint

Address: North Sydney, NSW, 2060

Date Founded: 11/01/2025

Web Address: <https://www.holdpoint.com/>

Number of Employees: 1FT 1PT

Founding Team

Founders & Leadership

Name: Cameron Harris

Title: Founder & CEO

Email: cameron@holdpoint.com

Name:

Title:

Email:

Funding & Traction

Revenue

2024: N/A

2025: N/A

2026 proj.: Pre-revenue

Latest Round of Funding

Bootstrapped

Angel

Pre Seed

Seed

Series A

Series B

Grants

DoE Funded

Funding Raised to Date:

Grants, Contracts, Memberships

Pilot with Kaptol Group: Australia's largest Data Center builder in active development. DPR Construction pilot meeting occurred on 6/5 and that has progressed to the next stage of selecting project specific.

TAM, Tech, & Supplemental

Total Addressable Market

\$500 million

Tech Readiness Level

①

②

③

④

⑤

⑥

⑦

⑧

⑨

Supplemental Documents

Company Description

Problem Statement & Target Customer

Mission-critical infrastructure projects, data centers, BESS, substations, lose \$200-500K per day when energization is delayed. The blocker isn't the work: it's proving the work is done. QA evidence lives across SharePoint folders, emails, and spreadsheets. No single system tracks what is complete, what is outstanding, and what is actually blocking power-on. Ideal customer: electrical subcontractor completions managers and PMs on projects above \$30M with card-swipe authority and direct QA pain.

Value Proposition

Holdpoint is the single source of truth between construction complete and power-on. It maps every cable to its inspection records and test results, giving PMs a live energization readiness view across the entire site, without changing how field teams work today. The result: fewer delays, stronger progress claim substantiation, and a clear, auditable path to energization. "Proof before power."

Product or Service

Holdpoint is an energization readiness platform. Contractors upload a cable schedule, tag critical path assets, and invite subcontractors to link existing QA evidence directly to each asset. The platform shows real-time completion status, flags what is blocking energization, generates progress reports for claim substantiation, and produces a full audit log for handover. V1 sits on top of SharePoint.

Founding Team Description

15 years as a licensed electrician and project engineer delivering mission-critical infrastructure across APAC. Personally experienced a \$480K progress payment withheld due to inability to prove QA/QC % completion, the exact problem Holdpoint solves rapidly. Dual US/AU citizen with active networks across the Aus/US construction markets. Domain depth is the moat: this product was designed by someone who has held the cable schedule, chased the test sheets, and stood in front of the energy marshal.

Technologies & Unique Know-How

Holdpoint's core tech is a cable-keyed compliance ledger: an asset-agnostic data structure that links any doc, photo, or test result to a specific asset and stage-gates energization clearance at the circuit level. SharePoint authentication enables zero-disruption adoption, no new document system required. The ledger is trade-agnostic by design, making electrical the wedge and multi-trade completions the expansion path. No competitor combines a cable schedule ledger with an energization gate.

Traction & Competitive Landscape

Pilot discussions advanced stage w/ Tier 1 Builders in Aus, and DPR and Devcon in the US. Active partnership MOU executed with Hardhat Robotics(US). API partnership in dev with Fieldwire and Bluebeam. Competitors task completion (Conqa, Visibuild, Sitemate) confirm checklists done but cannot clear a circuit for energization, and commissioning records platforms (CxAlloy, BlueRithm) equipment-specific. Holdpoint sits above both. No existing tool answers: is this circuit or site ready to energize?

Pilot Demonstration & Cost

A pilot runs on a single active project 20-40% complete, before QA/QC ramps up. An ideal partner is an electrical subcontractor or general contractor building a data center, BESS, hospital or substation project above \$20M. Cameron sets up the project: uploads the cable schedule, tags critical path assets, invites the sub. The sub links existing QA evidence to each asset in SharePoint. The GC gets a live energization readiness dashboard. Duration: 8-10 weeks. One project with measurable output. Cost: \$2000

Company Overview

Point of Contact Information

Name: Yann Pfitzer
Title: Founder & CEO
Email: yann@kanopy-ai.com
Phone: (267) 449-9837

Company Information

Legal Name: KanopyAI
Address: 33 W 17th Street, Ste 6, New York, NY, 10011
Date Founded:
Web Address: <https://www.kanopy-ai.com/>
Number of Employees: 2FT

Founding Team

Founders & Leadership

Name: Yann Pfitzer
Title: CEO
Email: yann@kanopy-ai.com



Name: Nicolas Pfitzer
Title: CTO
Email: nicolas@kanopy-ai.com

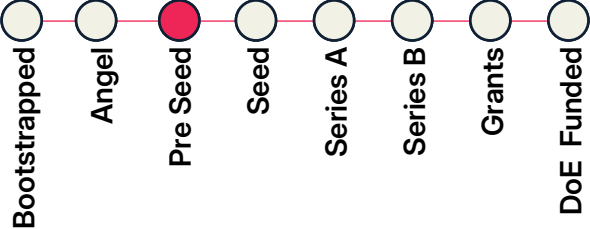


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$155,000

Latest Round of Funding



Funding Raised to Date: \$500,000

Grants, Contracts, Memberships

Two commercial contracts in our first three months from launch: a top-10 investor-owned utility and a national contractor partner, totaling \$155K committed 2026 ARR and 3,000 paid inspection miles.
Antler

TAM, Tech, & Supplemental

Total Addressable Market

\$18 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Utilities operate millions of miles of grid using asset and vegetation data that is often months to years old. This drives reactive work, mis-allocated O&M spend, outage risks, and manual inspection burden. Our ideal customer is any electric utility or UVM contractor managing thousands of overhead line miles, with recurring vegetation, asset inspection, compliance, emergency response, and work-planning needs

Value Proposition

Kanopy turns ordinary drone or vehicle video into a live, measurable 3D model of the grid. We help utilities and contractors automatically identify vegetation and asset risks earlier, prioritize work based on field conditions, reduce manual inspection and QA/QC effort, and create records at a fraction of the cost and turnaround time of LiDAR or satellite data

Product or Service

Kanopy sells a computer-vision platform for utility field intelligence. All we need is ordinary video - no LiDAR, sensors, or specialized hardware. Customers upload drone, vehicle, or field footage, and our models turn it into a measurable 3D twin of the grid: detecting vegetation and asset conditions, measuring clearances, assigning risk scores, and generating work scopes, maps, reports, and compliance records to help utilities plan, prioritize, dispatch, and audit field work.

Founding Team Description

Kanopy is led by brothers Yann and Nicolas Pfitzer. Yann was an Associate Partner in McKinsey's energy practice, where he led strategy, growth, and business-building work for utilities, energy companies, and infrastructure investors. Nicolas is a robotics and machine-learning engineer with an MSc from ETH Zurich, research experience at Cambridge, and prior experience building computer-vision systems at ShapeScale.

Technologies & Unique Know-How

Kanopy's core technology is a feed-forward computer-vision pipeline that turns ordinary video into 3D grid intelligence without calibration, LiDAR, sensors, or specialized hardware. We have trained our own models to infer 3D structure, detect vegetation and grid assets, measure clearances, and score risk directly from footage. This lets us deliver LiDAR-like intelligence with much faster turnaround at ~80% lower cost.

Traction & Competitive Landscape

Kanopy has \$155K in committed 2026 revenue so far, including a paid pilot with a top-10 utility and a 7-year contract with a national utility services firm. We are engaged with 20+ utilities, contractors, and strategics representing 3M miles of T&D. Competitors rely on LiDAR or satellite; Kanopy delivers LiDAR-like outputs, with faster turnaround, lower cost, and vegetation plus asset insights.

Pilot Demonstration & Cost

A small pilot would cover 100 miles of overhead distribution lines. The utility or contractor captures ordinary vehicle, drone, UTV, or mobile video, or Kanopy can collect it. Kanopy processes the footage into 3D reconstructions, tree-level clearance measurements, health/species flags, span-level vegetation metrics, risk-ranked work scopes, and platform access for review. Ideal partners are utilities or UVM contractors with active inspection, planning, compliance, or post-work audit needs. Cost: \$1,000

TIRION SYSTEMS



Company Overview

Point of Contact Information

Name: David Trejo
Title: CEO
Email: dt@tirion.systems
Phone: +5491162345733

Company Information

Legal Name: Tirion Systems
Address: Buenos Aires, CABA, 1186

Date Founded: 01/05/2026
Web Address: <https://tirion.systems/>
Number of Employees: 3FT 1PT

Company Description

Problem Statement & Target Customer

100k utilities with <100k connections run blind, using paper logs and WhatsApp. They manage 30% of global distribution capacity but cannot afford \$500k+ enterprise software. Our ideal customer is an electric cooperative or municipal utility with 10k-100k meters, 1-3 IT staff, and a \$40k software budget.

Value Proposition

We deliver physical grid observability in 30 days using the utility's existing billing and GIS data, requiring zero new hardware to start. We replace enterprise deployments with an affordable SaaS subscription that small utility boards can approve in one meeting.

Product or Service

We sell Ulmo Grid, a physical observability SaaS. It features a pre-built utility ontology mapping network assets, an AI agent layer for automated complaint triage and regulatory reporting, and optional \$180 sensors. We charge an annual subscription per active connection.

Founding Team Description

I am a 3x founder with two exits (including Nydro Energy to Aramco) and the former National Director of Energy Transition of Argentina. Our team includes utility network veterans with 10+ years at Siemens. We combine deep-tech AI expertise with actual grid operation experience.

Technologies & Unique Know-How

Our core tech is a pre-built utility ontology allowing LLMs to reason about physical grids like code. We built a connector library to ingest messy legacy data. We also developed our own \$180 ESP32/LoRa sensors, creating a strong dual software-hardware integration moat.

Traction & Competitive Landscape

We have a live \$150k contract with a 60k-meter utility (Tandil) and 5 more onboarding (~\$700k ARR path by Q3). 45 utilities are in pipeline. Competitors like Palantir or Siemens take 18+ months and \$1M+. Lower-tier SCADA vendors lack AI.

Pilot Demonstration & Cost

A 30-day deployment integrating the utility's existing billing, GIS, and complaint data. Within 4 weeks, operators receive a live energy balance dashboard and an AI complaint triage agent. Zero new hardware is required to start. Our ideal partner is a U.S. rural electric cooperative (NRECA) or municipal utility with 10k-60k meters operating on legacy systems.

Cost: \$15,000 - \$25,000

Founding Team

Founders & Leadership

Name: David Trejo
Title: CEO
Email: dt@tirion.systems



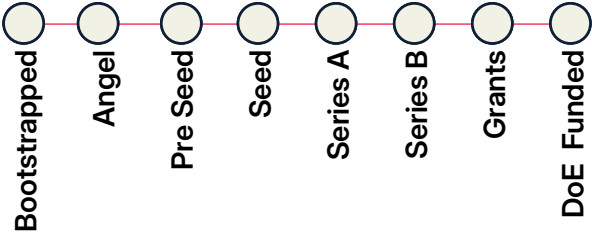
Name:
Title:
Email:

Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$150,000

Latest Round of Funding



Funding Raised to Date:

Grants, Contracts, Memberships

Live \$150k contract with Tandil (60k meters), negotiating a 10-year master agreement. 5 more utilities are onboarding, tracking to \$700k ARR by Q3 2026.

TAM, Tech, & Supplemental

Total Addressable Market

\$1.5 billion

Tech Readiness Level



Supplemental Documents



OMNISPECTRA



Company Overview

Point of Contact Information

Name: Joseph Schmidt
Title: Cofounder
Email: joseph.schmidt@omnispectra.ai
Phone:

Company Information

Legal Name: OmniSpectra
Address: 3723 Greenville Ave STE 33861,
Dallas, TX 75206
Date Founded: 4/20/2026
Web Address: <https://omnispectra.netlify.app/>
Number of Employees:

Founding Team

Founders & Leadership

Name: Joseph Schmidt
Title: Cofounder
Email: joseph.schmidt@omnispectra.ai

Name: Kanav Saraf
Title: Cofounder
Email: kanav.saraf@omnispectra.ai

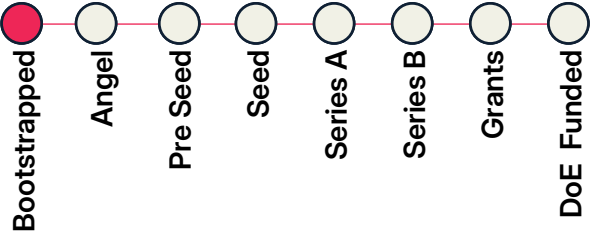


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: N/A

Latest Round of Funding



Funding Raised to Date: \$10,000

Grants, Contracts, Memberships

NSF I-Corps, UT Austin SEED, and TEX-E; Multiple major utilities have expressed interest in field pilots following successful testbed validation. 100+ customer discovery interviews across utilities, OEMs, and the broader ecosystem; UT Austin Forty Acres MIT Engine Blueprint, Q-Station Energy Accelerator

TAM, Tech, & Supplemental

Total Addressable Market

\$30 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Utilities face aging transformer fleets, growing maintenance backlogs, and limited field crews, making it difficult to identify which assets require attention before failures occur. Existing continuous monitoring solutions are often too expensive to deploy fleet-wide. Our ideal customers are investor-owned utilities, public power utilities, cooperatives, and other grid operators seeking to reduce O&M costs, prioritize inspections, extend transformer life, and improve grid reliability.

Value Proposition

OmniSpectra enables utilities to shift from reactive maintenance to proactive asset prioritization. Our affordable, non-invasive sensing platform continuously identifies which transformers need attention next week, next month, or next year - reducing unnecessary inspections, preventing costly failures, extending asset life, and helping utilities maximize existing maintenance budgets without disrupting established workflows.

Product or Service

OmniSpectra delivers a hardware, software, and decision-support platform for continuous transformer health monitoring. Our non-invasive sensor installs without modifying the transformer and feeds actionable health insights through flexible software integrations or utility workflows. Rather than replacing existing diagnostics, OmniSpectra helps utilities determine where to focus inspections and advanced testing first.

Founding Team Description

Founded by two PhDs with 15+ years of combined experience in sensor technology and startups, OmniSpectra combines deep technical expertise with rigorous customer discovery. Supported by NSF I-Corps, The Engine (MIT), and The University of Texas at Austin, our team has conducted 100+ interviews across the electric power ecosystem to ensure we're building technology utilities want to deploy.

Technologies & Unique Know-How

OmniSpectra has developed proprietary, non-invasive sensing technology adapted from advanced aerospace inspection methods for power grid applications. Our platform continuously measures transformer health without intrusive installation, enabling affordable fleet-scale deployment where existing monitoring solutions are cost-prohibitive. Our focus is actionable maintenance prioritization - not simply fault detection.

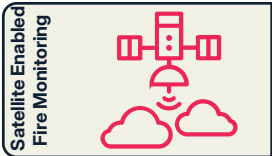
Traction & Competitive Landscape

OmniSpectra is validating its technology through leading startup and energy innovation programs, including NSF I-Corps, The Engine (MIT), TEX-E, and Q-Station. Incumbent OEMs and utilities have engaged with us as we prepare for testbed validation. Unlike high-cost monitoring systems deployed on a small fraction of transformers or fault-detection solutions that react late, OmniSpectra enables affordable, fleet-scale maintenance prioritization to reduce O&M costs and extend asset life.

Pilot Demonstration & Cost

A pilot would deploy OmniSpectra sensors on a small group of substation transformers (e.g., 5-10 assets) to continuously monitor health, prioritize maintenance, and validate performance against existing utility inspection methods. Ideal partners include investor-owned utilities, public power utilities, cooperatives, or utility testbeds seeking to reduce O&M costs, improve asset management, and accelerate predictive maintenance. Cost: \$75,000 - \$150,000, depending on the number of assets, duration, and reporting and integration needs.

ORBITAL SENTRY



Company Overview

Point of Contact Information

Name: Elizabeth Foughty
Title: Chief Business Officer
Email: elizabeth@orbitalsentry.ai
Phone: (703) 795-4785

Company Information

Legal Name: Orbital Sentry
Address: 80 M St SE, Washington, DC 20003
Date Founded: 11/25/2025
Web Address: <https://www.orbitalsentry.ai/>
Number of Employees: 2FT 2PT

Founding Team

Founders & Leadership

Name: Mark Keremedjiev
Title: CEO
Email: mark@orbitalsentry.ai



Name: Elizabeth Foughty
Title: Chief Business Officer
Email: elizabeth@orbitalsentry.ai

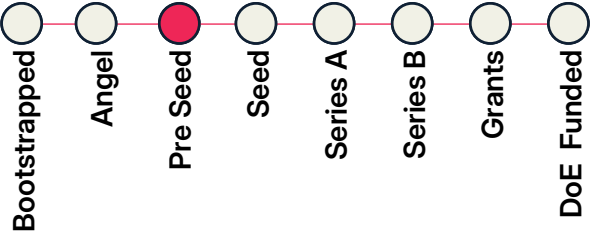


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: Confidential

Latest Round of Funding



Funding Raised to Date: \$2,000,000

Grants, Contracts, Memberships

Delos, Property Guardian, The Hartford, Seneca Drones, Wildfire Defense Systems, Major Western Utility (under NDA); InnSure

TAM, Tech, & Supplemental

Total Addressable Market

\$5 billion

Tech Readiness Level

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

Supplemental Documents



Company Description

Problem Statement & Target Customer

Wildfires are one of the greatest threats to grid resilience. Utility transmission lines are a leading ignition source, and today's leading sensing technologies are too slow to catch a fire while it's still stoppable. Utilities resort to costly, blunt-force power shutoffs because they can't see risk in real time. Fire Sentry will deliver continuous infrared monitoring from geosynchronous orbit with sub-15-second refresh. Ideal customers are US utilities, insurers, and government fire agencies.

Value Proposition

No existing satellite delivers continuous infrared coverage. LEO systems pass overhead periodically. GOES-R refreshes every 10 minutes at 2 km resolution, designed for weather, not fire. Fire Sentry will be the first commercial geosynchronous platform purpose-built for wildfire detection, with sub-15-second data refresh and sensitivity down to 1 square meter. Persistent coverage means fires caught at ignition, not after they've grown into grid-threatening disasters.

Product or Service

Orbital Sentry will sell continuous infrared monitoring data from geosynchronous orbit, delivered as real-time alerts, fire progression feeds, and AI-ready time-series datasets. Before launch, we are generating high-fidelity synthetic data for partners to integrate into existing risk models and workflows today. Post-launch, commercially exposed organizations (utilities, insurers, reinsurers) will fund access through a consortium model, while first responders receive key data elements for free.

Founding Team Description

Our team has done this before, across every layer of the stack. CEO Dr. Mark Keremedjiev led satellite missions at Planet Labs and ran \$500M+ government programs. CBO Elizabeth Foughty commercialized satellite data at Planet and spent a decade selling AI and geospatial products to the world's largest insurers.

Technologies & Unique Know-How

Fire Sentry is built on three core technologies: fire-optimized short-wave and midwave infrared sensing tuned for thermal anomaly detection; geosynchronous orbit providing continuous, staring coverage impossible from LEO; and AI change-detection algorithms trained on persistent infrared image stacks. The combination produces something no existing system offers: fires detected at 1 square meter, with sub-15-second data refresh, 24/7.

Traction & Competitive Landscape

No existing system offers continuous infrared coverage. GOES-R refreshes every 10 minutes at 2 km resolution and is designed for weather, not fire. LEO constellations produce periodic snapshots, not persistent monitoring. Airborne and ground sensors are expensive and geo limited. Fire Sentry fills that gap. We have commercial commitments from insurers, utilities, and fire response organizations, with endorsements from Watch Duty's CEO and San Bernardino County's Fire Chief.

Pilot Demonstration & Cost

We will have 2 varieties of pilot. Initially, 1) A synthetic data program that will allow users to test data pre-launch and later 2) a pathfinder program that will provide lower resolution satellite data. An ideal partner is a utility or platform company that will benefit from continuous wildfire monitoring data. This will probably be within in a team that manages PSPS or wildfire risk. Cost: 1) \$15,000 2) \$150,000

FIRESCAPE



Fire Risk Mitigation

GRID PROTECTION

Company Overview

Point of Contact Information

Name: Holly Eagleston
Title: CEO and Founder
Email: holly@firescape.ai
Phone: (207) 595-3018

Company Information

Legal Name: Firescape
Address: 424 Tulane Dr SE, Albuquerque, NM 87106
Date Founded: 10/08/2024
Web Address: <https://www.firescape.ai/>
Number of Employees: 3FT 7PT

Founding Team

Founders & Leadership

Name: Holly Eagleston
Title: CEO
Email: holly@firescape.ai



Name: Braden Smith
Title: CTO
Email: braden@firescape.ai

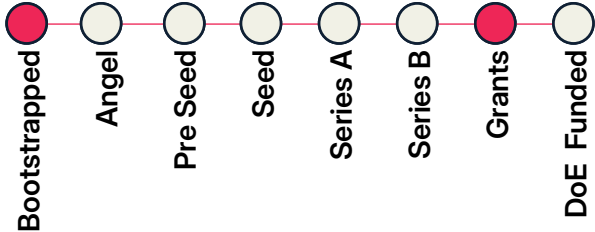


Funding & Traction

Revenue

2024:
2025: \$150,000
2026 proj.: \$500,000

Latest Round of Funding



Funding Raised to Date: \$1,500,000

Grants, Contracts, Memberships

NM EDD, NOAA SBIR
We have two pilots with investor owned utilities, and two contracts with rural coops.
NM LEEP

TAM, Tech, & Supplemental

Total Addressable Market

\$10 billion

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

We help utilities prevent electric-grid caused wildfire. We are purpose-built for utilities where wildfire is a new threat or have small internal teams and need a decision making tool that recommends actionable insights. Ideally, we would work with operations, planning or wildfire teams at a utility where wildfire is an emerging threat.

Value Proposition

We bring together siloed datasets, advanced fire spread modeling and weather data to help utilities balance safety with reliability. Our recommendations are actionable, and granular, allowing utilities to only do protection settings (like non-reclose) when they absolutely need to, for the smallest footprint. For planning, we help utilities determine the best ROI on resilience investments.

Product or Service

We provide a decision making tool, not more data, to forecast when to do protection settings. This provides granular insights to balance safety and reliability during high fire weather conditions, grounded in wildfire spread consequence models. For long-term planning, we quantify risk exposed and risk reduced by each mitigation strategy (covered conductors, replacing fuses, etc) and model the most effective cap-ex investment strategy.

Founding Team Description

Both founders come from Sandia National Labs, Dr. Holly Eagleston was a Principal Member of Technical Staff for 6 years where she lead projects in wildfire risk and energy security, before that she worked at the Bureau of Land Management mapping wildfire behavior. She has deep technical experience in the wildfire and geospatial science space, and together with Braden Smith, who has data engineering and software development experience, they have the technical skills to bring the product to market

Technologies & Unique Know-How

We use machine learning to predict when to do a protection setting based on in-house models of fuel moisture, wind speed, relative humidity and consequence of a wildfire. This builds off of work the founders did while at Sandia National Labs.

Traction & Competitive Landscape

Over the past year, we had 4 paid pilots, 2 of which have converted to sales. Our main competitors include Cloudfire, Pyrologix and Technosylva. We fill the gap between modeling reports and expensive simulation tools to help utilities make fast, repeatable decisions. We provide an end-to-end software tool for wildfire mitigation with email and text alerts for situational awareness.

Pilot Demonstration & Cost

An ideal partner would be a utility like Duke Energy who has emerging wildfire risk. For a pilot, we would set up our models for their region to tailor thresholds for protection settings for their territory. We would provide daily email and text alerts about the wildfire risk score and what recommended protection actions to implement for each circuit or protection device. They could see fire spread models and supporting data in our platform. We would need data from them to get started (GIS). Cost: \$100,000

Company Overview

Point of Contact Information

Name: Morgan Campbell
Title: Co-Founder & COO
Email: morgan@gridiq.io
Phone: (313) 720-7681

Company Information

Legal Name: GridIQ
Address: 518 Morgan Territory Road, Clayton, CA 94517
Date Founded: 7/10/2025
Web Address: <https://www.gridiq.io/>
Number of Employees: 3FT 1PT

Founding Team

Founders & Leadership

Name: Aren Page
Title: CEO
Email: aren@gridiq.io



Name: Morgan Campbell
Title: COO
Email: morgan@gridiq.io

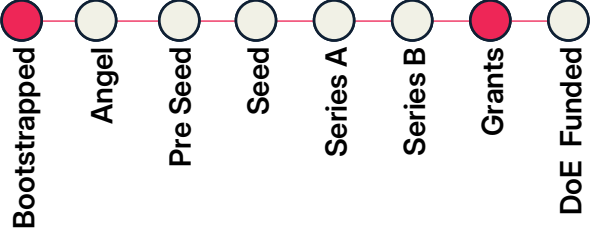


Funding & Traction

Revenue

2024: N/A
2025: N/A
2026 proj.: \$25,000

Latest Round of Funding



Funding Raised to Date: \$115,000

Grants, Contracts, Memberships

Startup Humboldt; We are currently negotiating a 7 figure deployment for early next year; LACI - current cohort

TAM, Tech, & Supplemental

Total Addressable Market

\$500 million

Tech Readiness Level



Supplemental Documents



Company Description

Problem Statement & Target Customer

Utilities are being asked to improve reliability and harden aging distribution grids while many faults are still found through outage calls, patrols, visual inspection, and truck rolls across miles of line. The result is slower restoration, higher O&M cost, greater crew exposure, and limited early warning. Our ideal customer is an IOU or municipal utility with medium-voltage feeders, reliability penalties, wildfire/storm risk, aging overhead assets, and budget for grid modernization.

Value Proposition

Our platform helps utilities see where faults are emerging or occurred, dispatch crews to the right span faster, reduce patrol time, improve SAIDI/SAIFI, and make better wildfire, resilience, and asset-management decisions in both energized & de-energized environments. We give operators actionable line-health intelligence without replacing existing SCADA/protection systems or rebuilding the grid, creating a scalable path from pilot to systemwide visibility.

Product or Service

GridIQ sells a hardware-enabled SaaS platform for medium-voltage distribution diagnostics. Compact field devices install along utility circuits and send controlled diagnostic signals through the line. Our software analyzes reflections and anomalies, localizes faults, classifies likely failure modes, and delivers alerts through dashboards, APIs, and control-room workflows.

Founding Team Description

CEO Aren Page brings firsthand utility distribution experience from PG&E, including wildfire hardening, undergrounding, substations, and grid modernization. CTO Josh Sweere holds an MS in EE from Penn and brings RF/signal-processing depth from Lockheed Martin plus hardware commercialization at First Mode, where he helped grow the company to a ~\$1.5B valuation. COO Morgan Campbell has scaled software partnership & GTM programs at Google Cloud, Attentive etc before co-founding SF Climate Week.

Technologies & Unique Know-How

GridIQ's patent-pending core technology is active grid diagnostics: controlled line interrogation, reflectometry-inspired signal analysis, rugged medium-voltage hardware, and ML models that classify anomalies over time. Unlike cameras, acoustic devices, or traditional protection systems, GridIQ is designed to detect and localize electrical faults directly on distribution infrastructure. Our edge is combining hardware, signal processing, and utility workflows from field device to control room.

Traction & Competitive Landscape

GridIQ is pre-commercial with strong early traction: a signed LOI with Blue Lake Rancheria, planned testing with Idaho National Laboratory, and utility pilot conversations through LACI/ LADWP. Competitors include visual wildfire detection, acoustic pole sensors, and EMF sensing technologies. GridIQ's unique approach provides greater data & context over time, before a fault occurs, and operates on energized & de-energized lines.

Pilot Demonstration & Cost

Our ideal pilot is a deployment on an energized medium-voltage distribution feeder with an electric cooperative or utility partner. We are seeking a minimum 0.5km test segment –and ideally 5 km section of overhead distribution line where we can validate continuous monitoring, fault detection, and precise fault localization under real operating conditions. The ability to induce different fault types would be ideal for demonstrating fault classification. Cost: GridIQ has funds available to support pilot deployments that meaningfully advance our TRL.

With Special Thanks

To our outstanding Research Associates

Joules Accelerator is incredibly fortunate to have such a deep bench of student leaders and mission driven young professionals. The ARL to Scale program, Future Grid Network, and Cohort 17 search were all made possible by this group.

Thank you!



Aditi Vikram
Spring '26 Research Associate

Duke University
B.S. Economics, Minor in Environmental Sciences & Policy



Jake Schwartz
Spring '26 Research Associate

University of North Carolina at Chapel Hill
Master's degree, Information Science- Class of 2026



Miles Blackhart
Spring '26 Research Associate

Duke University
B.S. Economics and Environmental Sciences/Policy



Aaryansh Vaish
Spring '26 Research Associate

Duke University Fuqua School of Business
Master of Quantitative Management, MQM- Class of 2026



Lauren Beck
Spring '26 Research Associate

Duke University Nicholas School of the Environment | Fuqua School of Business
MEM/MBA Candidate - Class of 2027



Hannah Stark
Spring '26 Research Associate

Duke University
B.E. Mechanical Engineering



Mallika Subramanian
Summer '26 Research Associate

Purdue University
B.S. Environmental and Ecological Engineering



About Joules Accelerator

Joules Accelerator is a 501(c)(3) nonprofit startup accelerator dedicated to advancing grid innovations and accelerating the energy transition. By connecting utilities, industry leaders, investors, municipalities, and universities, Joules reduces commercialization risk for early-stage startups and helps them move from concept to deployment to broader adoption.

Through programs like the ARL to Scale commercialization framework and the newly launched Future Grid Network, Joules creates pilot opportunities, fosters cross-sector collaboration, and builds a pipeline of talent and solutions that strengthen the energy ecosystem across the Carolinas and beyond.

Contact

info@joulesaccelerator.com

Follow us on

